
Bayesian Inference

Detailed, Self-Explanatory Solution Manual

Coverage.

This manual solves all 45 problems in the supplied practice set, including the three integrated challenge problems. For every problem the presentation is:

Exact Assignment Question $\rightarrow$ **Theory Required** $\rightarrow$ **Step-by-Step Derivation** $\rightarrow$
Interpretation

The theory sections are source-grounded in the supplied Bayesian lecture notes and reproduce the definitions, identities, and formulas needed to solve the corresponding problem. Minimal connective explanations are added where necessary so that every displayed step explains why the next step follows.

Prepared directly from the supplied assignment and lecture-note files.

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How to Read the Solutions

The supplied notes emphasize one recurring Bayesian mechanism:

$$\pi(\theta | x) \propto \pi(\theta)f(x | \theta)$$

and organize Bayesian analysis around constructing the probability model, conditioning on observed data, and checking model fit. The practice set likewise asks the reader repeatedly to identify the model, prior, likelihood, posterior, and resulting inference or prediction.

Throughout this manual, the algebra is deliberately not compressed. Whenever a transformation is used, the reason is stated; whenever a term is discarded under proportionality, it is identified as being constant with respect to the unknown parameter; and whenever a probability statement changes meaning, the random quantity is explicitly identified.

Core workflow from the supplied notes

$$\text{Model} \longrightarrow \text{Prior} \longrightarrow \text{Likelihood} \longrightarrow \text{Posterior} \longrightarrow \text{Inference / Prediction}$$

The most important identity is

$$\pi(\theta | x) \propto \pi(\theta)f(x | \theta).$$

For conjugate models, the main computational task is to recognize the resulting kernel. Powers of θ and $1 - \theta$ signal a Beta kernel; a quadratic exponential in a Normal mean signals a Normal kernel after completing the square.

For model comparison,

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}.$$

For hierarchical models, observations can be conditionally independent given a latent parameter while becoming dependent after that parameter is integrated out.

Chapter 1

Classical and Bayesian Foundations

1.1 Question 1.1 — The Meaning of Uncertainty

Exact question from the assignment:

Suppose

$$X_1, \dots, X_n \stackrel{\text{ind}}{\sim} N(\theta, 1).$$

After observing the data, a researcher obtains

$$\bar{X} = 2.3, \quad n = 100.$$

- (a) Construct the classical 95% confidence interval for θ .
- (b) Explain precisely why the statement

$$P(\theta \in CI \mid X) = 0.95$$

is not a valid frequentist interpretation.

- (c) Now suppose a Bayesian analysis uses

$$\theta \sim N(0, 100).$$

Derive the posterior distribution.

- (d) Construct a 95% posterior credible interval.
- (e) Explain carefully why the Bayesian credible interval permits a probability statement about θ after observing the data, whereas the classical confidence interval does not.
- (f) As $n \rightarrow \infty$, compare the role played by the prior in the Bayesian analysis with the role played by repeated sampling in the frequentist interpretation.

Theory required before solving

For a normal sampling model

$$X_i \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2), \quad \sigma^2 \text{ known,}$$

the sample mean satisfies

$$\bar{X} \sim N\left(\theta, \frac{\sigma^2}{n}\right).$$

A classical 95% confidence interval is obtained from the pivotal quantity

$$\frac{\bar{X} - \theta}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Its 95% coverage statement is a repeated-sampling statement:

$$\Pr_{\theta}\{L(X) \leq \theta \leq U(X)\} = 0.95.$$

After the observed data are fixed, the interval endpoints are fixed numbers and θ is a fixed but unknown constant.

In Bayesian inference, the parameter is assigned a prior distribution and the posterior is

$$\pi(\theta | x) \propto \pi(\theta)L(\theta; x).$$

A 95% credible interval is a posterior-probability statement:

$$\Pr(\theta \in [L, U] | x) = 0.95.$$

Step-by-step solution**(a) Classical 95% confidence interval.**

Here $\sigma^2 = 1$ and $n = 100$, so

$$\text{SE}(\bar{X}) = \frac{1}{\sqrt{100}} = 0.1.$$

For a two-sided 95% interval, use $z_{0.975} = 1.96$. Therefore

$$\bar{X} \pm 1.96(0.1) = 2.3 \pm 0.196.$$

Hence

$$\boxed{CI_{0.95} = (2.104, 2.496)}.$$

(b) Why $P(\theta \in CI | X) = 0.95$ is not the frequentist interpretation.

The interval produced by the procedure is

$$CI(X) = \left[\bar{X} - 1.96/\sqrt{n}, \bar{X} + 1.96/\sqrt{n} \right].$$

Before sampling, $CI(X)$ is random because it depends on the random sample X . By

construction,

$$\Pr_{\theta}\{\theta \in CI(X)\} = 0.95.$$

After observing $\bar{X} = 2.3$, the interval becomes the fixed numerical interval $(2.104, 2.496)$. In the frequentist framework, θ is not assigned a probability distribution. Thus it is not meaningful within that framework to say that the already-fixed interval contains the fixed parameter with conditional probability 0.95.

(c) Posterior distribution.

The prior is

$$\theta \sim N(0, 100),$$

so

$$\mu_0 = 0, \quad \tau_0^2 = 100.$$

The normal-normal conjugacy formula gives

$$\tau_1^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \right)^{-1} = \left(\frac{1}{100} + 100 \right)^{-1}.$$

Since

$$\frac{1}{100} + 100 = 100.01 = \frac{10001}{100},$$

we obtain

$$\tau_1^2 = \frac{100}{10001} \approx 0.009999.$$

The posterior mean is

$$\mu_1 = \frac{\mu_0/\tau_0^2 + n\bar{X}/\sigma^2}{1/\tau_0^2 + n/\sigma^2} = \frac{0 + 100(2.3)}{0.01 + 100}.$$

Therefore

$$\mu_1 \approx 2.29977.$$

Hence

$$\theta | X \sim N(2.29977, 0.009999).$$

Why this happens: the data have precision $n/\sigma^2 = 100$, while the prior has precision $1/100 = 0.01$. The data are therefore overwhelmingly more precise.

(d) Bayesian 95% credible interval.

The posterior standard deviation is

$$\sqrt{\tau_1^2} \approx 0.099995.$$

Thus

$$\mu_1 \pm 1.96\sqrt{\tau_1^2} \approx 2.29977 \pm 0.19599.$$

Hence

$$CrI_{0.95} \approx (2.10378, 2.49576).$$

(e) Why the interpretations differ.

The Bayesian interval satisfies

$$\Pr(2.10378 < \theta < 2.49576 \mid X) = 0.95.$$

The randomness here comes from the posterior distribution of θ , conditional on the observed data.

The frequentist interval instead means that if the whole experiment were repeated indefinitely and the same interval-construction rule were applied each time, then 95% of those random intervals would contain the fixed true value of θ .

(f) What happens as $n \rightarrow \infty$?

For the Bayesian model,

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \right)^{-1} \rightarrow 0,$$

and

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{X}}{1/\tau_0^2 + n/\sigma^2} \rightarrow \bar{X}.$$

Under the law of large numbers, $\bar{X} \rightarrow \theta_{\text{true}}$, so the posterior concentrates around the truth and the effect of the fixed prior becomes negligible.

The frequentist interval also shrinks because its half-width is

$$1.96 \frac{\sigma}{\sqrt{n}} \rightarrow 0.$$

Thus the two approaches can become numerically very similar for large samples, although their probability interpretations remain different.

1.2 Question 1.2 — Statistical vs Practical Significance

Exact question from the assignment:

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1).$$

A scientist believes that effects satisfying

$$|\theta| < 0.01$$

are scientifically negligible.

Assume that

$$\theta = 0.002$$

and

$$n = 10^8.$$

- (a) Compute approximately the classical test statistic for testing

$$H_0 : \theta = 0 \quad \text{vs} \quad H_1 : \theta \neq 0.$$

- (b) Determine whether the classical test at level 0.05 would reject H_0 .
 (c) Explain why rejection of H_0 does not imply practical importance.
 (d) Suppose instead that the Bayesian quantity of interest is

$$P(|\theta| < 0.01 \mid X).$$

Explain why this question is more directly aligned with the scientific objective.

- (e) Give an example of a decision rule based on posterior probabilities that distinguishes practical significance from statistical significance.

Theory required before solving

For $X_i \stackrel{\text{iid}}{\sim} N(\theta, 1)$,

$$\bar{X} \sim N\left(\theta, \frac{1}{n}\right).$$

Testing $H_0 : \theta = 0$ uses

$$Z = \frac{\bar{X} - 0}{1/\sqrt{n}} = \sqrt{n} \bar{X}.$$

A level-0.05 two-sided test rejects when $|Z| > 1.96$.

Statistical significance concerns compatibility with a null value under repeated sampling.

Practical significance instead asks whether the parameter is large enough to matter scien-

tifically. A Bayesian posterior permits direct questions such as

$$\Pr(|\theta| < 0.01 \mid X).$$

Step-by-step solution

(a) Classical test statistic.

The true parameter is stated to be

$$\theta = 0.002,$$

and

$$n = 10^8.$$

The standard error of $\bar{X}$ is

$$\frac{1}{\sqrt{n}} = \frac{1}{10^4} = 10^{-4}.$$

Thus

$$Z = \frac{0.002}{10^{-4}} = 20.$$

Therefore

$$Z = 20.$$

(b) Decision at level 0.05.

Because

$$|20| > 1.96,$$

the test rejects $H_0 : \theta = 0$.

$$\text{Reject } H_0.$$

(c) Why this does not imply practical importance.

The scientific threshold says effects with

$$|\theta| < 0.01$$

are negligible. But the actual effect is

$$|\theta| = 0.002 < 0.01.$$

So the effect is scientifically small even though the enormous sample size makes the standard error tiny:

$$\text{SE}(\bar{X}) = 10^{-4}.$$

The test statistic compares the effect to sampling noise:

$$\frac{0.002}{10^{-4}} = 20,$$

so a tiny effect can look statistically overwhelming when n is huge.

Key distinction:

small effect $\nrightarrow$ statistically insignificant

and, conversely,

statistically significant $\nrightarrow$ practically important.

(d) Why $P(|\theta| < 0.01 \mid X)$ is more aligned with the scientific question.

The scientist's question is literally whether the effect lies inside the scientifically negligible region

$$(-0.01, 0.01).$$

A posterior probability answers that question directly:

$$\Pr(|\theta| < 0.01 \mid X).$$

This is a probability about the parameter after observing the data, rather than a test of the artificial point null $\theta = 0$.

(e) Example Bayesian decision rule.

One possible rule is:

Declare the effect practically negligible if $\Pr(|\theta| < 0.01 \mid X) \geq 0.95$.

Otherwise declare the evidence insufficient to regard the effect as practically negligible.

The number 0.95 is a decision threshold and can be changed according to the consequences of the decision.

Why this is conceptually stronger: it separates the scientific threshold 0.01 from the evidence threshold 0.95. The first encodes what “important” means; the second encodes how much posterior certainty is required.

1.3 Question 1.3 — Random Sample Size

Exact question from the assignment:

Suppose the sample size N is determined by a preliminary coin toss:

$$N = \begin{cases} 2, & \text{with probability } 1/2, \\ 1000, & \text{with probability } 1/2. \end{cases}$$

Conditional on $N = n$,

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

The estimator is

$$\bar{X}_N.$$

(a) Compute

$$E(\bar{X}_N | N).$$

(b) Use the law of total variance to derive

$$\text{Var}(\bar{X}_N).$$

(c) Explain why this unconditional variance may be misleading for describing the precision of the estimator after the experiment has actually been performed.

(d) Suppose you observe that $N = 1000$. Which variance is relevant for inference about μ ?

(e) Discuss the connection between this example and the distinction between pre-data and post-data uncertainty.

Theory required before solving

The law of total expectation says

$$E(T) = E\{E(T | N)\}.$$

The law of total variance says

$$\text{Var}(T) = E\{\text{Var}(T | N)\} + \text{Var}\{E(T | N)\}.$$

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then

$$E(\bar{X}_n | N = n) = \mu, \quad \text{Var}(\bar{X}_n | N = n) = \frac{\sigma^2}{n}.$$

Step-by-step solution**(a) Conditional expectation.**

For every realized value $N = n$,

$$\bar{X}_N = \bar{X}_n.$$

Therefore

$$\mathbb{E}(\bar{X}_N \mid N = n) = \mu.$$

Equivalently,

$$\mathbb{E}(\bar{X}_N \mid N) = \mu.$$

The important point is that although the sample size is random, conditioning on N tells us exactly which sample mean we are talking about.

(b) Unconditional variance.

Apply total variance:

$$\text{Var}(\bar{X}_N) = \mathbb{E}[\text{Var}(\bar{X}_N \mid N)] + \text{Var}[\mathbb{E}(\bar{X}_N \mid N)].$$

The second term is

$$\text{Var}(\mu) = 0.$$

For the first term,

$$\text{Var}(\bar{X}_N \mid N) = \frac{\sigma^2}{N}.$$

Hence

$$\text{Var}(\bar{X}_N) = \sigma^2 \mathbb{E}\left[\frac{1}{N}\right].$$

Now

$$P(N = 2) = P(N = 1000) = \frac{1}{2},$$

so

$$\mathbb{E}\left[\frac{1}{N}\right] = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{1000} = \frac{1}{4} + \frac{1}{2000} = 0.2505.$$

Therefore

$$\text{Var}(\bar{X}_N) = 0.2505 \sigma^2.$$

(c) Why this unconditional variance can be misleading after the experiment.

Before the experiment, the sample size is uncertain. There is a substantial chance of obtaining only $N = 2$, whose variance is

$$\frac{\sigma^2}{2}.$$

There is also a chance of obtaining $N = 1000$, whose variance is

$$\frac{\sigma^2}{1000}.$$

The unconditional quantity averages over these two different experimental situations. After the experiment, however, N is observed. At that point there is no remaining uncertainty about the sample size, so averaging over the unobserved alternative $N = 2$ is inappropriate.

(d) Observing $N = 1000$.

The relevant variance is the conditional variance:

$$\text{Var}(\bar{X}_N \mid N = 1000) = \frac{\sigma^2}{1000}.$$

(e) Pre-data versus post-data uncertainty.

Before sampling, N is random, so we should consider the distribution of possible experiments. After sampling, N is known. Bayesian reasoning makes this conditioning viewpoint especially explicit: inference conditions on the information actually observed.

Thus

pre-data uncertainty averages over possible experiments,

whereas

post-data uncertainty conditions on the experiment that actually occurred.

1.4 Question 2.1 — Screening Test with Prior Sensitivity

Exact question from the assignment:

A disease has prevalence

$$P(D) = p.$$

A diagnostic test satisfies

$$P(T^+ | D) = 0.99, \quad P(T^+ | D^c) = 0.05.$$

(a) Derive a general expression for

$$P(D | T^+)$$

as a function of p .

(b) Evaluate this posterior probability when

$$p = 0.001, \quad p = 0.01, \quad p = 0.10, \quad p = 0.50.$$

(c) Find the value of p for which

$$P(D | T^+) = \frac{1}{2}.$$

(d) Explain why a highly accurate test can still have a low posterior probability of disease after a positive result.

(e) Suppose the patient receives two conditionally independent positive test results. Derive

$$P(D | T_1^+, T_2^+).$$

(f) Compare the effect of the second test with the effect of changing the prior prevalence.

Theory required before solving

Bayes' theorem is

$$P(D | T^+) = \frac{P(T^+ | D)P(D)}{P(T^+ | D)P(D) + P(T^+ | D^c)P(D^c)}.$$

With prevalence $p = P(D)$,

$$P(D^c) = 1 - p.$$

Conditional independence means that if two test results are conditionally independent given disease status, then

$$P(T_1^+, T_2^+ | D) = P(T_1^+ | D)P(T_2^+ | D),$$

and similarly under D^c .

Step-by-step solution

(a) General posterior after one positive test.

Substitute

$$P(T^+ | D) = 0.99, \quad P(T^+ | D^c) = 0.05.$$

Then

$$P(D | T^+) = \frac{0.99p}{0.99p + 0.05(1 - p)}.$$

Expanding the denominator,

$$0.99p + 0.05 - 0.05p = 0.94p + 0.05.$$

Hence

$$P(D | T^+) = \frac{0.99p}{0.94p + 0.05}.$$

Why the denominator contains both terms: a positive test can occur either because a truly diseased person tests positive or because a healthy person produces a false positive.

(b) Numerical values.

For $p = 0.001$,

$$P(D | T^+) = \frac{0.00099}{0.00099 + 0.04995} \approx 0.01943.$$

Thus

$$\boxed{1.94\%}.$$

For $p = 0.01$,

$$P(D | T^+) = \frac{0.0099}{0.0099 + 0.0495} = \frac{0.0099}{0.0594} \approx 0.1667.$$

Thus

$$\boxed{16.67\%}.$$

For $p = 0.10$,

$$P(D | T^+) = \frac{0.099}{0.099 + 0.045} = \frac{0.099}{0.144} \approx 0.6875.$$

Thus

$$\boxed{68.75\%}.$$

For $p = 0.50$,

$$P(D | T^+) = \frac{0.495}{0.495 + 0.025} = \frac{0.495}{0.52} \approx 0.951923.$$

Thus

$$\boxed{95.19\%}.$$

(c) Find p such that $P(D | T^+) = 1/2$.

Set

$$\frac{0.99p}{0.99p + 0.05(1 - p)} = \frac{1}{2}.$$

Cross-multiply:

$$1.98p = 0.99p + 0.05 - 0.05p = 0.94p + 0.05.$$

Hence

$$1.04p = 0.05,$$

so

$$p = \frac{0.05}{1.04} \approx 0.04808.$$

Thus a prevalence of about 4.81% is required for a positive test to make disease more likely than not.

(d) Why an accurate test can still give a low posterior probability.

When disease is rare, the population contains many healthy people. Even a small false-positive probability can therefore generate many false positives.

For example, at $p = 0.001$, among a very large population the number of healthy people is about 999 times the number of diseased people. The 5% false-positive rate applied to the huge healthy group can overwhelm the 99% true-positive rate applied to the tiny diseased group.

Therefore diagnostic accuracy and posterior probability are different concepts:

posterior probability depends on both test accuracy and prevalence.

(e) Two conditionally independent positives.

Assuming conditional independence,

$$P(T_1^+, T_2^+ | D) = 0.99^2,$$

and

$$P(T_1^+, T_2^+ | D^c) = 0.05^2.$$

Bayes' theorem gives

$$P(D | T_1^+, T_2^+) = \frac{0.99^2 p}{0.99^2 p + 0.05^2 (1 - p)}.$$

Thus

$$P(D | T_1^+, T_2^+) = \frac{0.9801p}{0.9801p + 0.0025(1 - p)}.$$

(f) Effect of the second test versus changing the prior.

Bayesian updating can be expressed in odds:

$$\frac{P(D | T^+)}{P(D^c | T^+)} = \frac{P(D)}{P(D^c)} \times \frac{0.99}{0.05}.$$

The likelihood ratio is 19.8. With two independent positives, the evidence multiplier becomes

$$19.8^2 = 392.04.$$

So each independent positive multiplies the odds by another factor of 19.8.

Changing the prevalence instead changes the starting odds. Hence:

prior prevalence sets the starting point; test results multiply the odds by evidence.

1.5 Question 2.2 — Sequential Bayesian Updating

Exact question from the assignment:

Suppose there are two hypotheses,

$$H_1, \quad H_2,$$

with prior probabilities

$$P(H_1) = 0.2, \quad P(H_2) = 0.8.$$

Three independent observations are collected. Their likelihood ratios are

$$\frac{p(y_1 | H_1)}{p(y_1 | H_2)} = 5,$$

$$\frac{p(y_2 | H_1)}{p(y_2 | H_2)} = \frac{1}{4},$$

and

$$\frac{p(y_3 | H_1)}{p(y_3 | H_2)} = 10.$$

- (a) Compute the prior odds.
- (b) Update the odds after observing y_1 .
- (c) Update again after observing y_2 .
- (d) Update again after observing y_3 .
- (e) Compute the final posterior probability of H_1 .
- (f) Show that updating sequentially gives the same answer as using the joint likelihood of all three observations.
- (g) Explain why the posterior probability does not necessarily move monotonically toward one hypothesis as observations arrive.

Theory required before solving

For two hypotheses, posterior odds satisfy

$$\frac{P(H_1 | y)}{P(H_2 | y)} = \frac{P(H_1) p(y | H_1)}{P(H_2) p(y | H_2)}.$$

If observations are conditionally independent given the hypothesis, the joint likelihood

ratio factors:

$$\frac{p(y_1, y_2, y_3 \mid H_1)}{p(y_1, y_2, y_3 \mid H_2)} = \prod_{j=1}^3 \frac{p(y_j \mid H_1)}{p(y_j \mid H_2)}.$$

Step-by-step solution

(a) **Prior odds.**

$$\frac{P(H_1)}{P(H_2)} = \frac{0.2}{0.8} = \boxed{\frac{1}{4}}.$$

(b) **After y_1 .**

Multiply the prior odds by the first likelihood ratio:

$$O_1 = \frac{1}{4}(5) = \boxed{\frac{5}{4}} = 1.25.$$

(c) **After y_2 .**

Now use the posterior odds $5/4$ as the new prior odds:

$$O_2 = \frac{5}{4} \cdot \frac{1}{4} = \boxed{\frac{5}{16}} = 0.3125.$$

The evidence y_2 favors H_2 , because its likelihood ratio is below 1, so the odds move downward.

(d) **After y_3 .**

$$O_3 = \frac{5}{16} \cdot 10 = \boxed{\frac{25}{8}} = 3.125.$$

(e) **Final posterior probability.**

If posterior odds are

$$O_3 = \frac{P(H_1 \mid y)}{P(H_2 \mid y)},$$

then because the probabilities sum to one,

$$P(H_1 \mid y) = \frac{O_3}{1 + O_3}.$$

Thus

$$P(H_1 \mid y) = \frac{25/8}{1 + 25/8} = \frac{25}{33}.$$

Therefore

$$\boxed{P(H_1 \mid y) = \frac{25}{33} \approx 0.7576.}$$

(f) **Joint update gives the same result.**

The joint likelihood ratio is

$$5 \cdot \frac{1}{4} \cdot 10 = 12.5.$$

Thus

$$O_{\text{final}} = \frac{1}{4}(12.5) = 3.125 = \frac{25}{8},$$

which is exactly the sequential result.

So

sequential multiplication of evidence = one joint Bayesian update.

(g) Why the posterior need not move monotonically.

The sequence of odds is

$$0.25 \longrightarrow 1.25 \longrightarrow 0.3125 \longrightarrow 3.125.$$

The first observation favors H_1 , the second favors H_2 , and the third strongly favors H_1 . Bayesian updating does not require the posterior to move steadily in one direction. It responds to the direction and strength of each new piece of evidence:

posterior belief can move up, down, and up again as evidence changes.

1.6 Question 2.3 — Three-Prisoners Generalization

Exact question from the assignment:

Three prisoners A, B, C have exactly one pardon.

The prior probabilities are not necessarily equal:

$$P(A) = a, \quad P(B) = b, \quad P(C) = c,$$

where

$$a + b + c = 1.$$

Prisoner A asks the guard to name one prisoner among B and C who will be executed.

Suppose that:

- If A is pardoned, the guard names B with probability q and C with probability $1 - q$.
- If B is pardoned, the guard must name C .
- If C is pardoned, the guard must name B .

Let G_B denote the event that the guard names B .

(a) Derive

$$P(A \mid G_B)$$

in terms of a, b, c, q .

(b) Find the corresponding expression for

$$P(C \mid G_B).$$

(c) Determine the conditions under which observing G_B provides no information about whether A is pardoned.

(d) Specialize your answer to

$$a = b = c = \frac{1}{3}, \quad q = \frac{1}{2}.$$

(e) Explain why treating the guard's statement as if it were a random event independent of the pardon assignment gives the wrong answer.

Theory required before solving

Bayes' theorem for an event A and observed event G_B is

$$P(A | G_B) = \frac{P(G_B | A)P(A)}{P(G_B)}.$$

The denominator is obtained by summing over the mutually exclusive pardon hypotheses:

$$P(G_B) = P(G_B | A)P(A) + P(G_B | B)P(B) + P(G_B | C)P(C).$$

The likelihood $P(G_B | \text{hypothesis})$ matters because the guard's statement mechanism depends on who is pardoned.

Step-by-step solution

(a) Derive $P(A | G_B)$.

The guard names B with probability q if A is pardoned:

$$P(G_B | A) = q.$$

If B is pardoned, the guard must name C :

$$P(G_B | B) = 0.$$

If C is pardoned, the guard must name B :

$$P(G_B | C) = 1.$$

Therefore

$$P(G_B) = aq + 0 \cdot b + c = aq + c.$$

Bayes' theorem gives

$$P(A | G_B) = \frac{aq}{aq + c}.$$

(b) Derive $P(C | G_B)$.

Similarly,

$$P(C | G_B) = \frac{P(G_B | C)P(C)}{P(G_B)} = \frac{c}{aq + c}.$$

Hence

$$P(C | G_B) = \frac{c}{aq + c}.$$

As a check,

$$P(A | G_B) + P(C | G_B) = \frac{aq + c}{aq + c} = 1,$$

which is correct because B cannot be pardoned if the guard names B as the prisoner to be executed.

(c) When does G_B provide no information about whether A is pardoned?

“No information” means the posterior equals the prior:

$$P(A \mid G_B) = P(A) = a.$$

Assuming $a > 0$,

$$\frac{aq}{aq + c} = a.$$

Cross-multiply:

$$aq = a(aq + c).$$

Thus

$$aq = a^2q + ac,$$

so

$$aq(1 - a) = ac.$$

Since $1 - a = b + c$,

$$\boxed{q(b + c) = c}$$

is the condition.

Equivalently,

$$\boxed{q = \frac{c}{b + c}}$$

when $b + c > 0$.

There is also a trivial degenerate case $a = 0$, where A was impossible from the start.

(d) Equal priors and $q = 1/2$.

Set

$$a = b = c = \frac{1}{3}, \quad q = \frac{1}{2}.$$

Then

$$P(A \mid G_B) = \frac{(1/3)(1/2)}{(1/3)(1/2) + 1/3} = \frac{1/6}{1/2} = \boxed{\frac{1}{3}}.$$

Likewise,

$$P(C \mid G_B) = \frac{1/3}{1/2} = \boxed{\frac{2}{3}}.$$

Thus hearing the guard name B leaves A 's probability at $1/3$ in this particular randomized-guard mechanism, while C 's probability becomes $2/3$.

(e) Why treating G_B as independent is wrong.

If one incorrectly assumed

$$P(G_B \mid A) = P(G_B \mid B) = P(G_B \mid C),$$

then the guard's statement would carry no evidence at all, because the likelihood factor would be the same under every hypothesis.

But the actual mechanism is:

$$P(G_B \mid A) = q, \quad P(G_B \mid B) = 0, \quad P(G_B \mid C) = 1.$$

These likelihoods are different, so G_B is informative about the underlying pardon state. The correct Bayesian update must include this data-generating mechanism.

Bayesian conditioning requires the correct likelihood of the observed statement.

Chapter 2

Likelihood Principle and Stopping Rules

2.1 Question 3.1 — Two Experiments, One Likelihood

Exact question from the assignment:

Consider Bernoulli observations with success probability p .

Experiment A fixes

$$n = 20$$

in advance and observes

$$x = 14$$

successes.

Experiment B continues sampling until the 14th success occurs and observes that the total number of trials required is 20.

(a) Write the likelihood function $L_A(p)$.

(b) Write the likelihood function $L_B(p)$.

(c) Show that

$$L_A(p) \propto L_B(p).$$

(d) Under a prior

$$p \sim \text{Beta}(\alpha, \beta),$$

derive the posterior distribution under both experiments.

(e) Show that the two posterior distributions are identical.

(f) Explain why a frequentist analysis may nevertheless produce different p -values.

(g) Discuss whether the stopping rule should influence a Bayesian inference about p in this situation.

Theory required before solving

The likelihood principle says that, for Bayesian updating, the observed likelihood function as a function of the parameter contains the evidential information. If two experiments lead to likelihoods proportional in the parameter,

$$L_1(\theta) \propto L_2(\theta),$$

then multiplication by the same prior and normalization gives the same posterior.

For Bernoulli data with x successes and $n - x$ failures,

$$L(p) \propto p^x (1 - p)^{n-x}.$$

A Beta prior has kernel

$$\pi(p) \propto p^{\alpha-1} (1 - p)^{\beta-1}.$$

Therefore the posterior is

$$p \mid \text{data} \sim \text{Beta}(\alpha + x, \beta + n - x).$$

Step-by-step solution**(a) Experiment A likelihood.**

We observe exactly $x = 14$ successes in $n = 20$ fixed trials:

$$P(X = 14 \mid p) = \binom{20}{14} p^{14} (1 - p)^6.$$

The binomial coefficient does not depend on p , so

$$L_A(p) \propto p^{14} (1 - p)^6.$$

(b) Experiment B likelihood.

Sampling continues until the 14th success occurs, and the 14th success occurs on trial 20. Therefore the first 19 trials must contain 13 successes and 6 failures, followed by a success:

$$P(T = 20 \mid p) = \binom{19}{13} p^{13} (1 - p)^6 p.$$

Hence

$$L_B(p) = \binom{19}{13} p^{14} (1 - p)^6,$$

and therefore

$$L_B(p) \propto p^{14} (1 - p)^6.$$

(c) Proportionality.

Comparing the two:

$$L_A(p) = \binom{20}{14} p^{14} (1-p)^6,$$

$$L_B(p) = \binom{19}{13} p^{14} (1-p)^6.$$

Thus

$$L_B(p) = \frac{\binom{19}{13}}{\binom{20}{14}} L_A(p),$$

where the ratio is a constant independent of p . Therefore

$$L_A(p) \propto L_B(p).$$

(d) Posterior under Experiment A.

Using

$$\pi(p) \propto p^{\alpha-1} (1-p)^{\beta-1},$$

we obtain

$$\pi(p \mid A) \propto p^{\alpha-1} (1-p)^{\beta-1} p^{14} (1-p)^6.$$

Collect powers:

$$\pi(p \mid A) \propto p^{\alpha+14-1} (1-p)^{\beta+6-1}.$$

Hence

$$p \mid A \sim \text{Beta}(\alpha + 14, \beta + 6).$$

(e) Posterior under Experiment B.

Because its likelihood has exactly the same p -dependent kernel,

$$\pi(p \mid B) \propto p^{\alpha+14-1} (1-p)^{\beta+6-1}.$$

Hence

$$p \mid B \sim \text{Beta}(\alpha + 14, \beta + 6).$$

Therefore

$$\pi(p \mid A) = \pi(p \mid B).$$

(f) Why frequentist p -values can differ.

A frequentist p -value depends on the sampling distribution of a test statistic under the null. That sampling distribution depends on the full stopping rule and on which outcomes would have been counted as evidence under repeated experiments.

Experiment A and Experiment B have different sample spaces and different stopping mechanisms. Even though the observed likelihood functions are proportional, the repeated-sampling reference distributions need not be identical. Hence the resulting p -values can differ.

(g) Should the stopping rule influence Bayesian inference here?

For this likelihood-principle calculation, no: the observed likelihoods are proportional, so with the same prior they produce the same posterior.

The important qualification is that this assumes the probabilistic model correctly captures the experiment. If the stopping mechanism changes the actual likelihood or introduces an unmodeled selection mechanism, then the Bayesian likelihood itself may be different. Thus it is not that stopping rules are magically irrelevant; rather,

once the correct observed likelihood is fixed, proportional likelihoods give the same Bayesian posterior.

2.2 Question 3.2 — Constructing Proportional Likelihoods

Exact question from the assignment:

Suppose

$$X \sim \text{Poisson}(\lambda).$$

Consider two different experiments.

Experiment A: Observe $n = 5$ independent Poisson observations whose sum is

$$S = 12.$$

Experiment B: Observe Poisson events sequentially until the cumulative count reaches at least 12, and suppose the resulting observed likelihood is proportional to

$$\lambda^{12} e^{-5\lambda}.$$

- (a) Derive the likelihood under Experiment A.
- (b) Verify whether the likelihoods are proportional.
- (c) Suppose

$$\lambda \sim \text{Gamma}(\alpha, \beta)$$

under a shape–rate parameterization. Derive the posterior under Experiment A.

- (d) Derive the posterior under Experiment B.
- (e) Explain which parts of the data-generating mechanism are relevant for Bayesian updating.
- (f) Give a careful interpretation of the statement:

“The likelihood contains all information about λ relevant to inference.”

Theory required before solving

For independent Poisson observations

$$X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda),$$

the joint likelihood is

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \propto e^{-n\lambda} \lambda^{\sum_i x_i}.$$

Thus the sum $S = \sum_i X_i$ is sufficient for λ .

For a Gamma(shape-rate) prior,

$$\lambda \sim \text{Gamma}(\alpha, \beta), \quad \pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda},$$

the posterior is

$$\lambda \mid x \sim \text{Gamma}(\alpha + S, \beta + n).$$

Step-by-step solution

(a) Experiment A likelihood.

Let the five independent observations be $X_1, \dots, X_5$, with

$$S = \sum_{i=1}^5 X_i = 12.$$

Then

$$L_A(\lambda) = \prod_{i=1}^5 \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = e^{-5\lambda} \lambda^{12} \prod_{i=1}^5 \frac{1}{x_i!}.$$

The product of factorials is independent of λ , so

$$L_A(\lambda) \propto \lambda^{12} e^{-5\lambda}.$$

(b) Compare with Experiment B.

Experiment B is explicitly stated to yield a likelihood proportional to

$$\lambda^{12} e^{-5\lambda}.$$

Therefore

$$L_A(\lambda) \propto L_B(\lambda).$$

(c) Posterior under Experiment A.

The prior is

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}.$$

Multiply prior and likelihood:

$$\pi(\lambda \mid A) \propto \lambda^{\alpha-1} e^{-\beta\lambda} \lambda^{12} e^{-5\lambda}.$$

Collect powers and exponential terms:

$$\pi(\lambda \mid A) \propto \lambda^{\alpha+12-1} e^{-(\beta+5)\lambda}.$$

This is the Gamma(shape-rate) kernel, so

$$\lambda \mid A \sim \text{Gamma}(\alpha + 12, \beta + 5).$$

(d) Posterior under Experiment B.

Because its likelihood is proportional to the same function,

$$\pi(\lambda \mid B) \propto \lambda^{\alpha-1} e^{-\beta\lambda} \lambda^{12} e^{-5\lambda},$$

and hence

$$\lambda \mid B \sim \text{Gamma}(\alpha + 12, \beta + 5).$$

(e) Which parts of the mechanism matter?

Bayesian updating uses

$$\pi(\lambda \mid \text{data}) \propto \pi(\lambda) L(\lambda; \text{observed data}).$$

Any factor in the likelihood that is constant in λ disappears after normalization. Thus the entire data-generating story matters insofar as it determines the correct likelihood, but once two experiments generate proportional likelihoods, the proportionality constant does not affect the posterior.

(f) Interpretation of “the likelihood contains all information about λ relevant for inference.”

This means that, conditional on the same prior, all information from the observed experiment that changes relative plausibility between different values of λ is encoded in

$$L(\lambda; \text{data}).$$

For two parameter values λ_1, λ_2 , the likelihood ratio

$$\frac{L(\lambda_1)}{L(\lambda_2)}$$

measures how the observed data compare those two parameter values. A multiplicative factor common to all λ cancels in this ratio and therefore cannot affect Bayesian posterior odds.

Thus the statement should not be misread as saying the likelihood describes every scientific feature of the experiment. It means that, for inference on λ , the parameter-dependent evidence relevant to Bayes’ rule is contained in the likelihood.

2.3 Question 3.3 — A Stopping Rule Thought Experiment

Exact question from the assignment:

A researcher observes Bernoulli trials and chooses one of the following stopping rules:

1. Stop after 100 observations.
2. Stop after observing 60 successes.
3. Stop when a likelihood ratio exceeds 100.

Suppose all three experiments eventually produce a data sequence containing exactly

60 successes and 40 failures.

- (a) Write the likelihood for p based only on the observed sequence.
- (b) Show that the likelihood has the form

$$L(p) \propto p^{60}(1-p)^{40}.$$

- (c) Under

$$p \sim \text{Beta}(2, 2),$$

derive the posterior.

- (d) Derive the posterior predictive probability of success on the next trial.
- (e) Explain why the three stopping rules do not change the posterior under the likelihood principle.
- (f) Give one practical situation where ignoring the stopping rule could nevertheless be problematic because the probabilistic model itself has been incorrectly specified.

Theory required before solving

For Bernoulli observations, if a realized data sequence contains x successes and $n - x$ failures, the likelihood kernel is

$$L(p) \propto p^x(1-p)^{n-x}.$$

Under a Beta prior,

$$p \sim \text{Beta}(\alpha, \beta),$$

posterior conjugacy gives

$$p \mid x \sim \text{Beta}(\alpha + x, \beta + n - x).$$

The likelihood principle says proportional likelihoods lead to identical Bayesian posteriors.

Step-by-step solution

(a) Likelihood for the observed sequence.

We observe a sequence with

$$x = 60, \quad n - x = 40.$$

For the particular realized sequence, the probability is

$$p^{60}(1 - p)^{40}.$$

If only the total count is recorded rather than the ordering, a binomial coefficient also appears; that coefficient is constant in p . Therefore in either representation,

$$L(p) \propto p^{60}(1 - p)^{40}.$$

(b) Required form.

Starting from the Bernoulli product,

$$L(p) = \prod_{i=1}^{100} p^{y_i} (1 - p)^{1 - y_i}.$$

Combine exponents:

$$L(p) = p^{\sum_i y_i} (1 - p)^{100 - \sum_i y_i}.$$

Since $\sum_i y_i = 60$,

$$L(p) \propto p^{60}(1 - p)^{40}.$$

(c) Posterior under $\text{Beta}(2, 2)$.

The prior kernel is

$$\pi(p) \propto p^{2-1}(1 - p)^{2-1} = p(1 - p).$$

Multiply by the likelihood:

$$\pi(p \mid y) \propto p(1 - p)p^{60}(1 - p)^{40} = p^{61}(1 - p)^{41}.$$

Match with

$$p^{a-1}(1 - p)^{b-1}.$$

Therefore

$$a = 62, \quad b = 42.$$

Hence

$$p \mid y \sim \text{Beta}(62, 42).$$

(d) Posterior predictive probability.

For a Beta posterior,

$$P(Y_{101} = 1 \mid y) = E[p \mid y].$$

The mean of $\text{Beta}(62, 42)$ is

$$\frac{62}{62 + 42} = \frac{62}{104} = \left[\frac{31}{52} \approx 0.59615 \right].$$

(e) Why stopping rules do not change the posterior here.

All three stopping rules eventually produce the same observed evidence: 60 successes and 40 failures. Thus the parameter-dependent likelihood kernel is the same:

$$p^{60}(1-p)^{40}.$$

Because the posterior is

$$\pi(p \mid y) \propto \pi(p)L(p; y),$$

the three proportional likelihoods result in the same posterior.

(f) When ignoring stopping could become problematic.

Suppose the researcher actually changes the sampling mechanism in a way that creates selection not represented by the assumed Bernoulli model. For example, imagine that after observing a streak of unusual outcomes the researcher selectively records only some trials, or that the success probability itself changes over time due to the stopping procedure.

Then the model

$$Y_i \mid p \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$$

is false. The problem is not the stopping rule as such; the problem is model misspecification. Bayesian updating is only as valid as the joint probability model used to construct the likelihood.

Chapter 3

Bayesian Modeling and Dependence

3.1 Question 4.1 — Building a Full Probability Model

Exact question from the assignment:

A factory produces electronic components.

For each component,

$$Y_i = \begin{cases} 1, & \text{defective,} \\ 0, & \text{not defective.} \end{cases}$$

Assume

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

but the true defect probability θ is unknown.

Historical information suggests

$$\theta \sim \text{Beta}(2, 98).$$

A sample of 50 components contains 4 defective items.

- (a) Explicitly write the full joint probability model

$$p(\theta, y_1, \dots, y_{50}).$$

- (b) Identify the observable and unobservable quantities.
- (c) Derive the posterior distribution.
- (d) Compute the posterior mean.
- (e) Derive the posterior predictive probability that the next component is defective.
- (f) Describe a posterior predictive check that could be used to evaluate whether the Bernoulli model is plausible.
- (g) Explain why obtaining a posterior distribution is not the final step of Bayesian data analysis.

Theory required before solving

Bayesian analysis begins by specifying a full joint model and then conditioning on observed data:

$$p(\theta, y) = p(\theta) p(y \mid \theta).$$

For Bernoulli observations,

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

implies

$$p(y_1, \dots, y_n \mid \theta) = \theta^z (1 - \theta)^{n-z}, \quad z = \sum_i y_i.$$

With the Beta prior

$$\theta \sim \text{Beta}(\alpha, \beta),$$

conjugacy gives

$$\theta \mid y \sim \text{Beta}(\alpha + z, \beta + n - z),$$

with posterior mean

$$E(\theta \mid y) = \frac{\alpha + z}{\alpha + \beta + n}.$$

For a future Bernoulli outcome,

$$P(Y_{n+1} = 1 \mid y) = E(\theta \mid y).$$

A posterior predictive check compares replicated data from the fitted model with the observed data.

Step-by-step solution

Let $n = 50$, $z = 4$, and prior $\text{Beta}(2, 98)$.

(a) Full joint probability model.

The prior density is

$$p(\theta) = \frac{1}{B(2, 98)} \theta^1 (1 - \theta)^{97}, \quad 0 < \theta < 1.$$

Conditional on θ ,

$$p(y_1, \dots, y_{50} \mid \theta) = \prod_{i=1}^{50} \theta^{y_i} (1 - \theta)^{1-y_i} = \theta^4 (1 - \theta)^{46}$$

for the observed data. Therefore

$$p(\theta, y_1, \dots, y_{50}) = p(\theta) \prod_{i=1}^{50} \theta^{y_i} (1 - \theta)^{1-y_i}.$$

(b) Observable and unobservable quantities.

The observations

$$Y_1, \dots, Y_{50}$$

are observable. The parameter θ is unobservable and represents the underlying defect probability.

(c) Posterior.

By conjugacy,

$$\theta \mid y \sim \text{Beta}(2 + 4, 98 + 46).$$

Thus

$$\theta \mid y \sim \text{Beta}(6, 144).$$

(d) Posterior mean.

For $\text{Beta}(a, b)$,

$$E(\theta) = \frac{a}{a+b}.$$

Therefore

$$E(\theta | y) = \frac{6}{150} = \boxed{0.04}.$$

Notice that the posterior mean equals 4%, while the observed sample proportion is also $4/50 = 0.08$. Why are they different? Because the prior mean is

$$\frac{2}{100} = 0.02$$

and the prior contributes 100 effective units of prior concentration, so it pulls the posterior downward.

(e) Predictive probability.

Conditionally on θ ,

$$P(Y_{51} = 1 | \theta, y) = \theta.$$

Average over the posterior:

$$P(Y_{51} = 1 | y) = E[\theta | y] = \boxed{0.04}.$$

(f) Posterior predictive check.

One natural check is to generate replicated samples

$$Y_1^{\text{rep}}, \dots, Y_{50}^{\text{rep}}$$

from the posterior predictive distribution and compare the number of defects

$$Z^{\text{rep}} = \sum_i Y_i^{\text{rep}}$$

with the observed $z = 4$. A discrepancy such as

$$T(y) = \sum_i y_i$$

or a day-level clustering statistic can be examined. If the observed statistic lies in an extreme part of its posterior predictive distribution, the Bernoulli model may be inadequate.

(g) Why posterior calculation is not the final step.

A posterior can be mathematically well-defined even when the assumed model is scientifically poor. Bayesian data analysis therefore has three conceptual stages:

$$\boxed{\text{model specification} \rightarrow \text{conditioning} \rightarrow \text{model checking.}}$$

The posterior answers questions conditional on the model; posterior predictive checking asks whether that model can reproduce important features of the observed data.

3.2 Question 4.2 — Model Misspecification

Exact question from the assignment:

Suppose the previous factory data were collected over 10 different production days.

Within each day, the defect probability may differ.

A proposed model is

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

However, an alternative model is

$$Y_{ij} \mid \theta_j \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta_j),$$

where j indexes the production day.

- (a) Explain the difference between the two probability models.
- (b) Under the first model, what type of dependence structure is induced marginally after integrating out θ ?
- (c) Why might posterior predictive checks reveal that the first model is inadequate?
- (d) Suggest a meaningful discrepancy statistic for comparing observed and replicated data.
- (e) Explain how Bayesian modeling separates the assumptions about the data-generating mechanism from the computational procedure used to obtain the posterior.

Theory required before solving

Conditional independence means independence after conditioning on the latent parameter:

$$P(Y_1, \dots, Y_n \mid \theta) = \prod_i P(Y_i \mid \theta).$$

After integrating out θ , observations can become dependent because they share uncertainty about the same latent parameter.

Posterior predictive checking asks whether replicated data generated from the fitted model resemble the observed data in scientifically relevant ways. A discrepancy statistic $T(y)$ summarizes a feature of the data used for checking.

Step-by-step solution

- (a) **Difference between the models.**

The first model assumes one common defect probability:

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

Thus all observations share exactly the same θ .

The second model allows a different defect probability for each production day:

$$Y_{ij} \mid \theta_j \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta_j).$$

This permits day-to-day heterogeneity.

So the first model says

$$\theta_1 = \cdots = \theta_{10} = \theta,$$

whereas the second permits

$$\theta_1, \dots, \theta_{10}$$

to differ.

(b) Marginal dependence under the first model.

Conditionally,

$$Y_i \perp Y_j \mid \theta.$$

But marginally,

$$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta) > 0$$

for $i \neq j$, provided θ is genuinely random.

Thus integrating out the common latent parameter induces positive dependence between observations.

(c) Why posterior predictive checks may reveal inadequacy.

Suppose production-day heterogeneity is real. The one- θ model may generate data that are too homogeneous across days. Observed data might show several unusually high-defect days and several unusually low-defect days, while simulations from the single- θ model rarely produce that degree of between-day variation.

Then a posterior predictive check can reveal systematic lack of fit.

(d) A discrepancy statistic.

A useful statistic is the between-day variance of defect proportions:

$$T(y) = \frac{1}{10} \sum_{j=1}^{10} (\hat{p}_j - \bar{p})^2,$$

where $\hat{p}_j$ is the observed defect fraction on day j .

Generate replicated datasets from the fitted model and compare

$$T(y^{\text{rep}})$$

with

$$T(y^{\text{obs}}).$$

If $T(y^{\text{obs}})$ is much larger than typical replicated values, the common- θ model is suspicious.

(e) Modeling assumptions versus computation.

The probability model determines what the data mean:

$$\text{likelihood} + \text{prior} = \text{posterior model.}$$

Computation is a separate issue. Two different models may both be easy or both be computationally difficult; conversely, a scientifically appropriate model may require MCMC. Thus:

$$\text{model choice is a statistical assumption; sampling/integration is a computational method.}$$

One cannot fix an incorrect data-generating model merely by using a more sophisticated algorithm.

3.3 Question 4.3 — Prediction as Bayesian Inference

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

where σ^2 is known.

You place a prior

$$\theta \sim N(\mu_0, \tau_0^2).$$

(a) Write the joint model for

$$\theta, Y_1, \dots, Y_n, \tilde{Y}.$$

(b) Derive

$$p(\tilde{Y} \mid Y_1, \dots, Y_n)$$

using

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta, Y) p(\theta \mid Y) d\theta.$$

(c) Explain why

$$p(\tilde{Y} \mid \theta, Y) = p(\tilde{Y} \mid \theta).$$

(d) Compare the uncertainty in θ with the uncertainty in $\tilde{Y}$.

(e) Explain why even if the posterior variance of θ approaches zero, the predictive distribution for $\tilde{Y}$ still retains observational variance.

Theory required before solving

For a future observation in a hierarchical Bayesian model,

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta, Y) p(\theta \mid Y) d\theta.$$

Conditional independence gives

$$p(\tilde{Y} \mid \theta, Y) = p(\tilde{Y} \mid \theta).$$

Thus posterior prediction is obtained by averaging the sampling distribution of the future observation over posterior uncertainty in the parameter.

For the normal-normal model,

$$\theta \mid Y \sim N(\mu_n, \tau_n^2), \quad \tilde{Y} \mid \theta \sim N(\theta, \sigma^2),$$

the sum representation

$$\tilde{Y} = \theta + \varepsilon$$

with independent $\varepsilon \sim N(0, \sigma^2)$ gives

$$\tilde{Y} \mid Y \sim N(\mu_n, \tau_n^2 + \sigma^2).$$

Step-by-step solution

(a) Joint model.

The model is

$$\theta \sim N(\mu_0, \tau_0^2),$$

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2),$$

and

$$\tilde{Y} \mid \theta \sim N(\theta, \sigma^2).$$

Hence

$$p(\theta, Y_1, \dots, Y_n, \tilde{Y}) = p(\theta) \left[\prod_{i=1}^n p(Y_i \mid \theta) \right] p(\tilde{Y} \mid \theta).$$

(b) Derive the posterior predictive distribution.

Start from

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta, Y) p(\theta \mid Y) d\theta.$$

Because of conditional independence,

$$p(\tilde{Y} \mid \theta, Y) = p(\tilde{Y} \mid \theta).$$

Therefore

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta) p(\theta \mid Y) d\theta.$$

Now represent

$$\tilde{Y} = \theta + \varepsilon,$$

where

$$\theta \mid Y \sim N(\mu_n, \tau_n^2), \quad \varepsilon \sim N(0, \sigma^2),$$

independently conditional on Y . Therefore a sum of independent normals is normal:

$$\tilde{Y} \mid Y \sim N(\mu_n, \tau_n^2 + \sigma^2).$$

(c) Why does conditional independence hold?

Once θ is known, the old observations and future observation are generated independently from the same sampling mechanism. Therefore the old data contain no additional information about $\tilde{Y}$ beyond what is already conveyed by θ .

$$\tilde{Y} \perp Y \mid \theta.$$

(d) Compare uncertainty in θ and $\tilde{Y}$.

The posterior variance of the parameter is

$$\text{Var}(\theta \mid Y) = \tau_n^2.$$

The predictive variance is

$$\text{Var}(\tilde{Y} \mid Y) = \tau_n^2 + \sigma^2.$$

The extra σ^2 is irreducible observational noise.

(e) Even if $\tau_n^2 \rightarrow 0$, what remains?

As $n \rightarrow \infty$,

$$\tau_n^2 \rightarrow 0.$$

Hence

$$\text{Var}(\tilde{Y} \mid Y) \rightarrow \sigma^2.$$

Therefore

$$\tilde{Y} \mid Y \rightarrow N(\theta_{\text{true}}, \sigma^2)$$

in the usual asymptotic sense.

Interpretation.

Learning the parameter and predicting an individual random outcome are different tasks:

parameter uncertainty can vanish while observation noise remains.

Chapter 4

Posterior Odds and Bayesian Evidence

4.1 Question 5.1 — Prior Odds vs Evidence

Exact question from the assignment:

Consider

$$H_1 : \theta = \theta_1, \quad H_2 : \theta = \theta_2.$$

Suppose

$$\frac{P(H_1)}{P(H_2)} = \frac{1}{9}.$$

After observing data y ,

$$\frac{p(y | H_1)}{p(y | H_2)} = 50.$$

(a) Compute the posterior odds.

(b) Compute

$$P(H_1 | y).$$

(c) How large must the likelihood ratio be for

$$P(H_1 | y) > \frac{1}{2}?$$

(d) Explain how strong evidence from the data can overcome an initially unfavorable prior.

(e) Suppose instead that the prior odds are

$$\frac{1}{10^6}.$$

What likelihood ratio is required to obtain posterior odds greater than 1?

(f) Discuss the interpretation of extremely strong prior beliefs.

Theory required before solving

For two competing hypotheses,

$$\text{posterior odds} = \text{prior odds} \times \text{Bayes factor}.$$

That is,

$$\frac{P(H_1 | y)}{P(H_2 | y)} = \frac{P(H_1) p(y | H_1)}{P(H_2) p(y | H_2)}.$$

If posterior odds are O , then

$$P(H_1 | y) = \frac{O}{1 + O}.$$

To make H_1 more probable than H_2 , posterior odds must exceed 1.

Step-by-step solution**(a) Posterior odds.**

The prior odds are

$$\frac{P(H_1)}{P(H_2)} = \frac{1}{9},$$

and the likelihood ratio is 50. Hence

$$O_{\text{post}} = \frac{1}{9}(50) = \boxed{\frac{50}{9} \approx 5.556}.$$

(b) Posterior probability of H_1 .

Convert odds to probability:

$$P(H_1 | y) = \frac{50/9}{1 + 50/9} = \frac{50}{59}.$$

Thus

$$\boxed{P(H_1 | y) = \frac{50}{59} \approx 0.8475}.$$

The initially disfavored hypothesis becomes strongly favored because the data contribute a Bayes factor 50.

(c) How large must the likelihood ratio be for posterior probability $> 1/2$?

We need

$$P(H_1 | y) > \frac{1}{2}.$$

This is equivalent to

$$O_{\text{post}} > 1.$$

Since

$$O_{\text{post}} = \frac{1}{9}BF_{12},$$

we require

$$\frac{1}{9}BF_{12} > 1.$$

Thus

$$BF_{12} > 9.$$

(d) Strong evidence can overcome an unfavorable prior.

The prior odds $1/9$ say that before data, H_2 is nine times as probable as H_1 . But the observed data are fifty times more likely under H_1 :

$$BF_{12} = 50.$$

Multiplying gives

$$\frac{1}{9} \times 50 > 1.$$

Therefore the posterior reverses the prior preference.

This is the central meaning of Bayesian evidence:

prior odds encode starting belief; the Bayes factor encodes what the data add.

(e) Prior odds 10^{-6} .

Now

$$\frac{P(H_1)}{P(H_2)} = 10^{-6}.$$

For posterior odds > 1 ,

$$10^{-6} BF_{12} > 1.$$

Therefore

$$BF_{12} > 10^6.$$

So an extremely strong prior preference requires correspondingly enormous data evidence to overturn it.

(f) Interpretation of extremely strong prior beliefs.

A prior odds ratio of 10^{-6} says the analyst assigns one million-to-one odds against H_1 before seeing the current data. Such a prior can be scientifically appropriate if it reflects overwhelming previous evidence, but it can also make posterior conclusions highly sensitive to the prior.

The Bayesian framework does not automatically declare the prior wrong. Instead it makes the tradeoff transparent:

$$\log O_{\text{post}} = \log O_{\text{prior}} + \log BF.$$

A very negative prior log-odds requires a large positive log-evidence to reverse the conclusion.

4.2 Question 5.2 — Posterior Odds Consistency

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} f(\cdot \mid \theta^*),$$

where

$$\theta^* \in \{\theta_1, \theta_2\}.$$

Define

$$R_n = \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_2)}.$$

(a) Show that

$$\log R_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_2)}.$$

(b) Assume $\theta^* = \theta_1$. Apply the law of large numbers to

$$\frac{1}{n} \log R_n.$$

(c) Show that the limiting expectation can be expressed using Kullback–Leibler divergence.

(d) Under suitable identifiability conditions, explain why

$$R_n \rightarrow \infty$$

almost surely.

(e) Deduce the asymptotic behavior of the posterior odds.

(f) Repeat the argument when

$$\theta^* = \theta_2.$$

Theory required before solving

For conditionally iid observations,

$$p(Y_1, \dots, Y_n \mid \theta) = \prod_{i=1}^n f(Y_i \mid \theta).$$

Therefore likelihood ratios multiply, and their logarithms add:

$$\log \frac{p(Y_{1:n} \mid \theta_1)}{p(Y_{1:n} \mid \theta_2)} = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_2)}.$$

Under the strong law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n g(Y_i) \rightarrow E_{\theta^*}[g(Y)] \quad \text{a.s.}$$

The KL divergence is

$$D_{\text{KL}}(P\|Q) = E_P \left[\log \frac{p(X)}{q(X)} \right] \geq 0,$$

with strict positivity for distinct identifiable distributions.

Step-by-step solution

Let

$$R_n = \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_2)}.$$

(a) Expand $\log R_n$.

Conditional independence gives

$$p(Y_1, \dots, Y_n \mid \theta_j) = \prod_{i=1}^n f(Y_i \mid \theta_j).$$

Therefore

$$R_n = \prod_{i=1}^n \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_2)}.$$

Taking logs converts the product into a sum:

$$\log R_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_2)}.$$

(b) Assume $\theta^* = \theta_1$.

Divide by n :

$$\frac{1}{n} \log R_n = \frac{1}{n} \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_2)}.$$

Since the observations are iid under θ_1 , the strong law gives

$$\frac{1}{n} \log R_n \xrightarrow{a.s.} E_{\theta_1} \left[\log \frac{f(Y \mid \theta_1)}{f(Y \mid \theta_2)} \right].$$

(c) Identify the limit as KL divergence.

By definition,

$$D_{\text{KL}}(f_{\theta_1} \| f_{\theta_2}) = E_{\theta_1} \left[\log \frac{f(Y \mid \theta_1)}{f(Y \mid \theta_2)} \right].$$

Hence

$$\frac{1}{n} \log R_n \rightarrow D_{\text{KL}}(f_{\theta_1} \| f_{\theta_2}).$$

(d) Why does $R_n \rightarrow \infty$?

If the model is identifiable and $\theta_1 \neq \theta_2$, then

$$D_{\text{KL}}(f_{\theta_1} \| f_{\theta_2}) > 0.$$

Thus for large n ,

$$\log R_n \approx n D_{\text{KL}}(f_{\theta_1} \| f_{\theta_2}),$$

which tends to $+\infty$ linearly. Exponentiating,

$$R_n \rightarrow \infty \quad \text{a.s.}$$

(e) Posterior odds.

Bayesian posterior odds satisfy

$$\frac{P(\theta_1 | Y_{1:n})}{P(\theta_2 | Y_{1:n})} = \frac{P(\theta_1)}{P(\theta_2)} R_n.$$

If the prior odds are finite and positive and $R_n \rightarrow \infty$, then

$$\frac{P(\theta_1 | Y_{1:n})}{P(\theta_2 | Y_{1:n})} \rightarrow \infty.$$

Thus the posterior probability of θ_1 approaches 1.

(f) True parameter $\theta^* = \theta_2$.

Now the law of large numbers yields

$$\frac{1}{n} \log R_n \rightarrow E_{\theta_2} \left[\log \frac{f(Y | \theta_1)}{f(Y | \theta_2)} \right].$$

Reverse the ratio:

$$= -D_{\text{KL}}(f_{\theta_2} \| f_{\theta_1}) < 0.$$

Therefore

$$\log R_n \rightarrow -\infty, \quad R_n \rightarrow 0.$$

Hence posterior odds in favor of θ_1 vanish and the posterior favors θ_2 .

The likelihood ratio learns the truth exponentially on the log scale, at a rate governed by KL divergence.

4.3 Question 5.3 — Conflicting Evidence

Exact question from the assignment:

Suppose a model comparison begins with

$$P(H_1) = P(H_2) = \frac{1}{2}.$$

Three batches of data provide Bayes factors:

$$BF_{12}^{(1)} = 20, \quad BF_{12}^{(2)} = \frac{1}{5}, \quad BF_{12}^{(3)} = \frac{1}{2}.$$

- (a) Compute the posterior odds after all three batches.
- (b) Compute the posterior probability of H_1 .
- (c) Which batch contributes the strongest evidence?
- (d) Explain why a single strong piece of evidence can be offset by later evidence.
- (e) Show that

$$\log BF_{12}^{\text{total}} = \sum_{j=1}^3 \log BF_{12}^{(j)}.$$

- (f) Explain why the log-likelihood ratio is particularly useful for studying asymptotic Bayesian consistency.

Theory required before solving

Bayes factors from independent batches of data multiply:

$$BF_{12}^{\text{total}} = \prod_j BF_{12}^{(j)}.$$

Consequently log-Bayes-factors add:

$$\log BF_{12}^{\text{total}} = \sum_j \log BF_{12}^{(j)}.$$

With prior odds 1, posterior odds equal the total Bayes factor.

Step-by-step solution

(a) Posterior odds.

The prior odds are

$$\frac{P(H_1)}{P(H_2)} = 1.$$

Therefore the final posterior odds are

$$O_{\text{post}} = 1 \times 20 \times \frac{1}{5} \times \frac{1}{2}.$$

Compute:

$$20 \cdot \frac{1}{5} = 4,$$

and

$$4 \cdot \frac{1}{2} = 2.$$

Hence

$$O_{\text{post}} = 2.$$

(b) Posterior probability.

Convert odds to probability:

$$P(H_1 | y) = \frac{2}{1+2} = \frac{2}{3} \approx 0.6667.$$

(c) Strongest evidence.

Evidence can be measured by how far the Bayes factor is from 1, or equivalently by the magnitude of its log.

The three Bayes factors are

$$20, \quad 0.2, \quad 0.5.$$

The strongest in favor of H_1 is 20. The second and third favor H_2 , because both are below 1. In absolute log terms, the first is

$$\log 20 \approx 2.996,$$

the second is

$$\log(0.2) = -\log 5 \approx -1.609,$$

and the third is

$$\log(0.5) \approx -0.693.$$

So the first batch contributes the largest magnitude of evidence.

(d) Why can later evidence offset earlier evidence?

Bayesian updating is multiplicative:

$$O_{\text{final}} = O_0 \prod_j BF_j.$$

A Bayes factor below 1 divides the current odds. Thus later data can overturn or weaken an earlier conclusion.

(e) Additivity of log Bayes factors.

Take logs:

$$\log BF_{\text{total}} = \log \left(BF^{(1)} BF^{(2)} BF^{(3)} \right).$$

Using

$$\log(ab) = \log a + \log b,$$

we obtain

$$\log BF_{\text{total}} = \log BF^{(1)} + \log BF^{(2)} + \log BF^{(3)}.$$

(f) Why logs are useful asymptotically.

For many iid observations, the likelihood ratio is a product of per-observation ratios. Products can become astronomically large or small, whereas logs turn them into sums:

$$\log R_n = \sum_{i=1}^n Z_i.$$

Then the law of large numbers applies directly:

$$\frac{1}{n} \log R_n \rightarrow E(Z).$$

That limit is a KL divergence with a sign determined by which parameter generated the data. Hence the log scale exposes the linear accumulation of evidence.

Chapter 5

Conditional Independence and Exchangeability

5.1 Question 6.1 — Conditionally Independent but Marginally Dependent

Exact question from the assignment:

Suppose

$$\theta \sim \text{Beta}(\alpha, \beta),$$

and

$$Y_1, Y_2 \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

(a) Compute

$$P(Y_1 = 1).$$

(b) Compute

$$P(Y_1 = 1, Y_2 = 1).$$

(c) Determine

$$P(Y_2 = 1 \mid Y_1 = 1).$$

(d) Show that, in general,

$$P(Y_1 = 1, Y_2 = 1) \neq P(Y_1 = 1)P(Y_2 = 1).$$

(e) Compute

$$\text{Cov}(Y_1, Y_2).$$

(f) Explain intuitively why observing Y_1 changes your prediction about Y_2 even though the observations are independent conditional on θ .

Theory required before solving

If

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

then

$$E(Y_i \mid \theta) = \theta, \quad \text{Var}(Y_i \mid \theta) = \theta(1 - \theta).$$

The law of total expectation and total variance give

$$E(Y_i) = E_\theta[E(Y_i \mid \theta)],$$

$$\text{Var}(Y_i) = E[\text{Var}(Y_i \mid \theta)] + \text{Var}[E(Y_i \mid \theta)].$$

For distinct observations,

$$\text{Cov}(Y_i, Y_j) = E[\text{Cov}(Y_i, Y_j \mid \theta)] + \text{Cov}(E[Y_i \mid \theta], E[Y_j \mid \theta]).$$

Conditional independence makes the first term 0, leaving

$$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta).$$

Step-by-step solution

Let

$$\theta \sim \text{Beta}(\alpha, \beta).$$

(a) Compute $P(Y_1 = 1)$.

Condition on θ :

$$P(Y_1 = 1) = E[P(Y_1 = 1 \mid \theta)].$$

Since $Y_1 \mid \theta \sim \text{Bernoulli}(\theta)$,

$$P(Y_1 = 1 \mid \theta) = \theta.$$

Therefore

$$P(Y_1 = 1) = E(\theta).$$

For a Beta distribution,

$$E(\theta) = \frac{\alpha}{\alpha + \beta}.$$

Hence

$$P(Y_1 = 1) = \frac{\alpha}{\alpha + \beta}.$$

(b) Compute $P(Y_1 = 1, Y_2 = 1)$.

Conditionally,

$$P(Y_1 = 1, Y_2 = 1 \mid \theta) = \theta^2$$

because the two variables are conditionally independent. Therefore

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2).$$

For $\theta \sim \text{Beta}(\alpha, \beta)$,

$$E(\theta^2) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}.$$

Thus

$$P(Y_1 = 1, Y_2 = 1) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}.$$

(c) Compute $P(Y_2 = 1 | Y_1 = 1)$.

By the definition of conditional probability,

$$P(Y_2 = 1 | Y_1 = 1) = \frac{P(Y_1 = 1, Y_2 = 1)}{P(Y_1 = 1)}.$$

Substitute the previous results:

$$= \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)} \frac{\alpha + \beta}{\alpha}.$$

Cancel common factors:

$$P(Y_2 = 1 | Y_1 = 1) = \frac{\alpha + 1}{\alpha + \beta + 1}.$$

(d) Show marginal non-independence.

Independence would require

$$P(Y_1 = 1, Y_2 = 1) = P(Y_1 = 1)P(Y_2 = 1).$$

Since the variables are identically distributed,

$$P(Y_1 = 1)P(Y_2 = 1) = \left(\frac{\alpha}{\alpha + \beta} \right)^2.$$

But in general

$$\frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)} \neq \frac{\alpha^2}{(\alpha + \beta)^2}.$$

Therefore

$$Y_1, Y_2 \text{ are generally not independent marginally.}$$

(e) Covariance.

Use

$$\text{Cov}(Y_1, Y_2) = E[\text{Cov}(Y_1, Y_2 | \theta)] + \text{Cov}(E[Y_1 | \theta], E[Y_2 | \theta]).$$

The first term is zero by conditional independence. The second is

$$\text{Cov}(\theta, \theta) = \text{Var}(\theta).$$

For a Beta distribution,

$$\text{Var}(\theta) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

Hence

$$\text{Cov}(Y_1, Y_2) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

(f) Intuition.

The observations do not directly influence each other once θ is known. But before θ is known, observing $Y_1 = 1$ increases our posterior belief that θ is large. A larger θ makes $Y_2 = 1$ more likely.

So the dependence is induced by the shared latent parameter:

$$Y_1 \perp Y_2 \mid \theta \quad \text{but} \quad Y_1 \not\perp Y_2.$$

This distinction is one of the central ideas of hierarchical Bayesian modeling.

5.2 Question 6.2 — Exchangeable but Not iid

Exact question from the assignment:

An urn contains exactly one red ball and one blue ball.

Two balls are drawn without replacement.

Let

$$Y_i = \begin{cases} 1, & \text{red,} \\ 0, & \text{blue.} \end{cases}$$

- (a) Find the joint distribution of

$$(Y_1, Y_2).$$

- (b) Show that

$$(Y_1, Y_2)$$

is exchangeable.

- (c) Compute

$$P(Y_2 = 1).$$

- (d) Compute

$$P(Y_2 = 1 \mid Y_1 = 1).$$

- (e) Show that the observations are not independent.

- (f) Explain why this example shows

$$\text{iid} \Rightarrow \text{exchangeable}$$

but not necessarily the converse.

- (g) Discuss why finite exchangeability does not automatically imply the existence of an infinite iid-given- θ representation.

Theory required before solving

Exchangeability means invariance of the joint distribution under permutations:

$$P(Y_1 = y_1, Y_2 = y_2) = P(Y_1 = y_2, Y_2 = y_1).$$

Independence requires factorization:

$$P(Y_1 = y_1, Y_2 = y_2) = P(Y_1 = y_1)P(Y_2 = y_2).$$

Thus exchangeability is weaker than iid. De Finetti's representation theorem requires

infinite exchangeability (or infinite extendability in the relevant binary setting), not merely finite exchangeability.

Step-by-step solution

(a) Joint distribution.

There are exactly two possible outcomes:

$$P(1, 0) = \frac{1}{2}, \quad P(0, 1) = \frac{1}{2},$$

and

$$P(0, 0) = P(1, 1) = 0.$$

Therefore

	$Y_2 = 0$	$Y_2 = 1$
$Y_1 = 0$	0	1/2
$Y_1 = 1$	1/2	0

(b) Exchangeability.

Swapping the labels 1 and 2 maps

$$(1, 0) \leftrightarrow (0, 1).$$

Both have probability 1/2. The other two configurations both have probability 0 and remain unchanged. Hence

$$P(Y_1 = y_1, Y_2 = y_2) = P(Y_1 = y_2, Y_2 = y_1)$$

for every pair, so

$$(Y_1, Y_2) \text{ is exchangeable.}$$

(c) Marginal $P(Y_2 = 1)$.

The event $Y_2 = 1$ occurs only for $(Y_1, Y_2) = (0, 1)$:

$$P(Y_2 = 1) = \frac{1}{2}.$$

(d) Conditional probability.

If $Y_1 = 1$, there is one red ball and it has already been drawn. The second draw must be blue:

$$P(Y_2 = 1 \mid Y_1 = 1) = 0.$$

(e) Not independent.

If they were independent, we would have

$$P(Y_1 = 1, Y_2 = 1) = P(Y_1 = 1)P(Y_2 = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

But actually

$$P(Y_1 = 1, Y_2 = 1) = 0.$$

Therefore

$$Y_1, Y_2 \text{ are not independent.}$$

(f) iid implies exchangeable, but not conversely.

For iid variables,

$$P(y_1, \dots, y_n) = \prod_i f(y_i),$$

and permuting the observations only permutes the factors. The product remains unchanged. Therefore iid implies exchangeability.

This urn example is exchangeable but dependent, so the converse is false:

$$\text{iid} \Rightarrow \text{exchangeable}, \quad \text{exchangeable} \not\Rightarrow \text{iid.}$$

(g) Why finite exchangeability is insufficient.

A finite exchangeable distribution can have structural restrictions that make an infinite extension impossible. Here the sequence contains only one red and one blue draw. It cannot be extended to infinitely many observations while preserving the same exact distributional structure.

An iid-given- θ model would instead allow arbitrarily many future observations. Therefore de Finetti's mixture representation requires infinite extendability:

$$\text{finite exchangeability alone does not guarantee a latent-}\theta \text{ mixture representation.}$$

5.3 Question 6.3 — Predicting the Next Observation

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta).$$

After observing

$$z = \sum_{i=1}^n Y_i$$

successes:

(a) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(b) Derive

$$P(Y_{n+1} = 1 \mid \theta, Y_1, \dots, Y_n).$$

(c) Explain the difference between these two probabilities.

(d) Show explicitly that the past observations affect the predictive distribution only through the posterior distribution of θ .

(e) Under what conditions does

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n)$$

approach

$$\frac{z}{n}?$$

(f) Interpret this result as a form of Bayesian learning.

Theory required before solving

Under the Beta-Bernoulli model,

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta),$$

and after z successes,

$$\theta \mid Y_{1:n} \sim \text{Beta}(\alpha + z, \beta + n - z).$$

Posterior predictive probabilities integrate over the posterior:

$$P(Y_{n+1} = 1 \mid Y_{1:n}) = \int P(Y_{n+1} = 1 \mid \theta, Y_{1:n}) \pi(\theta \mid Y_{1:n}) \, d\theta.$$

Step-by-step solution**(a) Derive $P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n)$.**

By conditional independence,

$$P(Y_{n+1} = 1 \mid \theta, Y_{1:n}) = \theta.$$

Therefore

$$P(Y_{n+1} = 1 \mid Y_{1:n}) = E[\theta \mid Y_{1:n}].$$

The posterior is

$$\text{Beta}(\alpha + z, \beta + n - z),$$

so its mean is

$$P(Y_{n+1} = 1 \mid Y_{1:n}) = \frac{\alpha + z}{\alpha + \beta + n}.$$

(b) Conditional probability given θ .Once θ is known, the future observation does not depend on the past:

$$P(Y_{n+1} = 1 \mid \theta, Y_1, \dots, Y_n) = \theta.$$

(c) Difference between the two probabilities.The first quantity averages over uncertainty in θ :

$$P(Y_{n+1} = 1 \mid Y_{1:n}) = E[\theta \mid Y_{1:n}].$$

The second assumes θ is known:

$$P(Y_{n+1} = 1 \mid \theta, Y_{1:n}) = \theta.$$

Thus the difference is exactly parameter uncertainty.

(d) How do the past data affect prediction?

The chain is

$$Y_{1:n} \longrightarrow \pi(\theta \mid Y_{1:n}) \longrightarrow P(Y_{n+1} = 1 \mid Y_{1:n}).$$

Once the posterior is known, the raw data no longer need to enter separately. The data affect prediction through the posterior distribution of θ .**(e) When does the predictive probability approach z/n ?**

We have

$$\frac{\alpha + z}{\alpha + \beta + n} = \frac{\frac{\alpha + \beta}{n} \frac{\alpha}{\alpha + \beta} + \frac{z}{n}}{1 + \frac{\alpha + \beta}{n}}.$$

If α, β are fixed while $n \rightarrow \infty$, then

$$\frac{\alpha + \beta}{n} \rightarrow 0.$$

Hence

$$\frac{\alpha + z}{\alpha + \beta + n} \rightarrow \frac{z}{n}$$

whenever z/n has a stable limit.

(f) Bayesian learning interpretation.

Initially, prediction reflects prior uncertainty:

$$P(Y_1 = 1) = \frac{\alpha}{\alpha + \beta}.$$

As observations accumulate, the posterior adapts. The data contribution z and $n - z$ increasingly dominates the fixed prior parameters.

Therefore

Bayesian learning is the progressive replacement of prior uncertainty by data-informed posterior uncertainty.

Chapter 6

de Finetti's Representation

6.1 Question 7.1 — Beta Mixing Produces Exchangeability

Exact question from the assignment:

Suppose

$$\theta \sim \text{Beta}(\alpha, \beta),$$

and conditional on θ ,

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

(a) Derive

$$P(Y_1 = y_1, \dots, Y_n = y_n)$$

by integrating out θ .

(b) Show that the resulting expression depends on the data only through

$$z = \sum_{i=1}^n y_i.$$

(c) Deduce that the sequence is exchangeable.

(d) Show that the observations are not generally independent marginally.

(e) Derive the marginal distribution of

$$Z = \sum_{i=1}^n Y_i.$$

(f) Identify this distribution.

Theory required before solving

If

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

then for a binary vector $y = (y_1, \dots, y_n)$,

$$p(y \mid \theta) = \theta^{\sum_i y_i} (1 - \theta)^{n - \sum_i y_i}.$$

The marginal joint distribution is obtained by integrating out θ :

$$p(y) = \int p(y \mid \theta) \pi(\theta) \, d\theta.$$

For $\theta \sim \text{Beta}(\alpha, \beta)$,

$$\pi(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

The Beta function identity is

$$B(a, b) = \int_0^1 t^{a-1} (1 - t)^{b-1} \, dt.$$

Step-by-step solution

Let

$$z = \sum_{i=1}^n y_i.$$

(a) Derive the marginal joint distribution.

First write the conditional likelihood:

$$p(y_1, \dots, y_n \mid \theta) = \theta^z (1 - \theta)^{n-z}.$$

The Beta prior density is

$$\pi(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

Therefore

$$p(y_1, \dots, y_n) = \int_0^1 \theta^z (1 - \theta)^{n-z} \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)} \, d\theta.$$

Combine powers:

$$= \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha+z-1} (1 - \theta)^{\beta+n-z-1} \, d\theta.$$

Recognize the Beta function:

$$p(y_1, \dots, y_n) = \frac{B(\alpha + z, \beta + n - z)}{B(\alpha, \beta)}.$$

(b) Dependence only on z .

The expression contains $y_1, \dots, y_n$ only through

$$z = \sum_i y_i.$$

Therefore any two binary sequences with the same number of successes have the same probability.

(c) Exchangeability.

Permuting the observations leaves z unchanged. Hence

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)})$$

for any permutation π . Thus

the sequence is exchangeable.

(d) Why not generally independent?

For independent Bernoulli variables marginally we would have

$$p(y_1, \dots, y_n) = \prod_i P(Y_i = y_i).$$

But the common random θ induces dependence. For $i \neq j$,

$$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta),$$

which is positive unless the prior on θ is degenerate.

conditional iid does not imply marginal iid when θ is random.

(e) Marginal distribution of Z .

Conditionally,

$$Z \mid \theta \sim \text{Binomial}(n, \theta).$$

Thus

$$P(Z = z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)} d\theta.$$

Therefore

$$P(Z = z) = \binom{n}{z} \frac{B(\alpha + z, \beta + n - z)}{B(\alpha, \beta)}.$$

(f) Identify the distribution.

This is exactly the beta-binomial distribution:

$$Z \sim \text{BetaBinomial}(n, \alpha, \beta).$$

Thus the Beta prior mixes the Binomial sampling distribution into a beta-binomial marginal

distribution.

6.2 Question 7.2 — Infinite vs Finite Exchangeability

Exact question from the assignment:

Consider a finite binary sequence

$$(Y_1, Y_2)$$

defined by

$$P(Y_1 = 1, Y_2 = 0) = \frac{1}{2},$$

$$P(Y_1 = 0, Y_2 = 1) = \frac{1}{2},$$

with all other joint probabilities equal to zero.

- (a) Show that (Y_1, Y_2) is exchangeable.
- (b) Show that the variables cannot be represented as conditionally iid Bernoulli observations under any mixing distribution on θ .
- (c) Hint: compute

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2).$$
- (d) Explain why this provides a warning against applying de Finetti's theorem to an arbitrary finite exchangeable collection.
- (e) State the role played by infinite extendability in the theorem.

Theory required before solving

De Finetti's theorem for binary sequences requires infinite exchangeability (equivalently, an infinitely extendable exchangeable sequence in this context). If

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

then necessarily

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2).$$

Since $\theta^2 \geq 0$, this quantity cannot equal zero while $P(Y_1 = 1) = E(\theta) > 0$ unless the mixing distribution is degenerate in an incompatible way.

Step-by-step solution

(a) Show exchangeability.

The only positive-probability outcomes are

$$(1, 0) \quad \text{and} \quad (0, 1),$$

both with probability $1/2$. Swapping the coordinates interchanges those outcomes without changing their probabilities. Therefore

$$(Y_1, Y_2) \text{ is exchangeable.}$$

(b) Show no conditional iid representation exists.

Suppose, for contradiction, there were a mixing distribution for which

$$Y_1, Y_2 \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

Then

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2).$$

But the given distribution says

$$P(Y_1 = 1, Y_2 = 1) = 0.$$

Since

$$\theta^2 \geq 0,$$

the expectation $E(\theta^2) = 0$ implies

$$\theta = 0 \quad \text{almost surely.}$$

That would imply

$$P(Y_1 = 1) = E(\theta) = 0.$$

But from the given joint distribution,

$$P(Y_1 = 1) = P(1, 0) = \frac{1}{2}.$$

Contradiction. Hence

$$\text{no such mixing distribution exists.}$$

(c) Using the hint.

The hint gives

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2) = 0.$$

Therefore $\theta = 0$ almost surely, which contradicts

$$E(\theta) = P(Y_1 = 1) = \frac{1}{2}.$$

(d) Warning about finite exchangeability.

The example is exchangeable for two observations, yet it cannot be represented as iid Bernoulli given a latent parameter. Therefore one cannot apply de Finetti merely from the fact that a finite vector is exchangeable.

finite exchangeability is not enough.

(e) Role of infinite extendability.

De Finetti requires that the exchangeable sequence can be continued to arbitrarily many observations while preserving exchangeability. This stronger condition forces the mixture-of-iid structure.

So the theorem's logical structure is:

infinite exchangeability $\implies$ mixture of iid Bernoulli laws.

The finite two-observation example fails precisely because it does not satisfy the required infinite extension property.

6.3 Question 7.3 — Recovering a Mixing Distribution

Exact question from the assignment:

Suppose an exchangeable binary sequence satisfies

$$P(Y_1 = 1) = \frac{1}{2},$$

and

$$P(Y_1 = 1, Y_2 = 1) = \frac{1}{3}.$$

Assume that the sequence has a de Finetti representation with

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta).$$

(a) Show that

$$E(\theta) = \frac{1}{2}.$$

(b) Show that

$$E(\theta^2) = \frac{1}{3}.$$

(c) Compute

$$\text{Var}(\theta).$$

(d) Compute

$$\text{Cov}(Y_1, Y_2).$$

(e) Suppose $\theta \sim \text{Beta}(\alpha, \alpha)$. Determine α so that the above two moment conditions hold.

(f) Explain how dependence among observations provides information about uncertainty in the latent parameter θ .

Theory required before solving

Under a de Finetti representation,

$$Y_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta),$$

we have

$$E(Y_1) = E(\theta)$$

and, by conditional independence,

$$E(Y_1 Y_2 \mid \theta) = E(Y_1 \mid \theta) E(Y_2 \mid \theta) = \theta^2.$$

Thus

$$E(Y_1 Y_2) = E(\theta^2).$$

Also,

$$\text{Cov}(Y_1, Y_2) = \text{Var}(\theta).$$

For $\theta \sim \text{Beta}(\alpha, \beta)$,

$$E(\theta) = \frac{\alpha}{\alpha + \beta},$$

and for $\text{Beta}(\alpha, \alpha)$,

$$E(\theta) = \frac{1}{2}, \quad \text{Var}(\theta) = \frac{1}{4(2\alpha + 1)}.$$

Step-by-step solution

(a) First moment.

We are given

$$P(Y_1 = 1) = \frac{1}{2}.$$

But

$$P(Y_1 = 1) = E[P(Y_1 = 1 \mid \theta)] = E(\theta).$$

Hence

$$E(\theta) = \frac{1}{2}.$$

(b) Second moment.

Because

$$P(Y_1 = 1, Y_2 = 1) = E(\theta^2),$$

the given value $1/3$ implies

$$E(\theta^2) = \frac{1}{3}.$$

(c) Variance of θ .

Use

$$\text{Var}(\theta) = E(\theta^2) - [E(\theta)]^2.$$

Thus

$$\text{Var}(\theta) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}.$$

(d) Covariance of Y_1, Y_2 .

From conditional independence,

$$\text{Cov}(Y_1, Y_2) = \text{Var}(\theta).$$

Therefore

$$\text{Cov}(Y_1, Y_2) = \frac{1}{12}.$$

(e) Find α under $\text{Beta}(\alpha, \alpha)$.

For a symmetric Beta,

$$\text{Var}(\theta) = \frac{\alpha\alpha}{(2\alpha)^2(2\alpha+1)} = \frac{1}{4(2\alpha+1)}.$$

Set this equal to $1/12$:

$$\frac{1}{4(2\alpha+1)} = \frac{1}{12}.$$

Cross-multiply:

$$12 = 4(2\alpha+1).$$

Divide by 4:

$$3 = 2\alpha + 1.$$

Hence

$$2\alpha = 2, \quad \boxed{\alpha = 1}.$$

So the mixing distribution is

$$\boxed{\theta \sim \text{Beta}(1, 1) = U(0, 1)}.$$

A useful consistency check is

$$E(\theta^2) = \frac{\alpha(\alpha+1)}{(2\alpha)(2\alpha+1)}$$

with $\alpha = 1$:

$$E(\theta^2) = \frac{2}{6} = \frac{1}{3},$$

as required.

(f) Interpretation.

If θ were fixed, conditionally independent observations would have zero covariance. Positive covariance tells us that the observations share uncertainty about a latent success probability. Indeed,

$$\text{Cov}(Y_1, Y_2) = \text{Var}(\theta).$$

Therefore stronger marginal dependence corresponds to greater uncertainty in the latent parameter:

$$\boxed{\text{dependence among observations reveals heterogeneity/uncertainty in } \theta}.$$

Chapter 7

Conjugate Bayesian Models

7.1 Question 8.1 — Prior Information as Pseudo-Observations

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_{20} \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta),$$

and the observed number of successes is

$$z = 12.$$

Consider the following priors:

$$\text{Prior A: } \theta \sim \text{Beta}(1, 1),$$

$$\text{Prior B: } \theta \sim \text{Beta}(20, 20),$$

$$\text{Prior C: } \theta \sim \text{Beta}(100, 20).$$

- (a) Derive the posterior distribution under each prior.
- (b) Compute the posterior mean under each prior.
- (c) Compute the posterior predictive probability of success.
- (d) Compare the posterior variances qualitatively.
- (e) Interpret the prior parameters α and β as prior information about successes and failures.
- (f) Which prior has the strongest influence on the posterior?
- (g) Explain whether it is correct to call α and β literal historical observations.

Theory required before solving

For Beta-Bernoulli conjugacy,

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Z \mid \theta \sim \text{Binomial}(n, \theta),$$

gives

$$\theta \mid Z = z \sim \text{Beta}(\alpha + z, \beta + n - z).$$

Its posterior mean and variance are

$$E(\theta \mid z) = \frac{\alpha + z}{\alpha + \beta + n},$$

$$\text{Var}(\theta \mid z) = \frac{(\alpha + z)(\beta + n - z)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}.$$

The quantity $\alpha + \beta$ controls prior concentration, while

$$\frac{\alpha}{\alpha + \beta}$$

is the prior mean.

Step-by-step solution

Here

$$n = 20, \quad z = 12, \quad \hat{\theta}_{\text{MLE}} = \frac{12}{20} = 0.6.$$

(a) Posterior under each prior.

For Prior A,

$$\text{Beta}(1, 1) \implies \theta \mid z \sim \text{Beta}(13, 9).$$

For Prior B,

$$\text{Beta}(20, 20) \implies \theta \mid z \sim \text{Beta}(32, 28).$$

For Prior C,

$$\text{Beta}(100, 20) \implies \theta \mid z \sim \text{Beta}(112, 28).$$

The update simply adds observed successes to the first shape parameter and failures to the second.

(b) Posterior means.

Prior A:

$$E(\theta \mid z) = \frac{13}{22} \approx 0.5909.$$

Prior B:

$$E(\theta \mid z) = \frac{32}{60} = 0.5333.$$

Prior C:

$$E(\theta \mid z) = \frac{112}{140} = 0.8.$$

The reason Prior C remains high is immediately visible: its prior mean is

$$100/120 = 0.8333,$$

so it pulls the posterior strongly upward despite the observed sample proportion 0.6.

(c) Posterior predictive probability.

For Beta-Bernoulli,

$$P(Y_{\text{next}} = 1 \mid z) = E(\theta \mid z).$$

Hence the predictive probabilities are respectively

$$\boxed{0.5909, \quad 0.5333, \quad 0.8.}$$

(d) Compare posterior variances.

Using

$$\text{Var}(\theta \mid z) = \frac{ab}{(a+b)^2(a+b+1)},$$

we obtain:

Prior A:

$$\frac{13 \cdot 9}{22^2 \cdot 23} \approx \boxed{0.01056}.$$

Prior B:

$$\frac{32 \cdot 28}{60^2 \cdot 61} \approx \boxed{0.004083}.$$

Prior C:

$$\frac{112 \cdot 28}{140^2 \cdot 141} \approx \boxed{0.001075}.$$

Thus Prior C gives the most concentrated posterior, Prior A the least concentrated.

(e) Pseudo-observation interpretation.

The Beta prior can be interpreted algebraically as if it contributes prior counts:

$$\alpha - 1 \quad \text{successes}, \quad \beta - 1 \quad \text{failures}$$

under one common convention, or more loosely as prior effective counts governed by α, β .

The exact “number of observations” interpretation depends on the chosen convention.

More importantly,

$$\alpha + \beta$$

controls the strength of the prior relative to the observed sample size n . Here:

$$2, \quad 40, \quad 120$$

are the total prior concentrations.

(f) Strongest influence.

Prior C has total concentration 120, compared with only 20 actual observations. Therefore it has the strongest influence:

$$\boxed{\text{Prior C.}}$$

(g) Are α, β literal historical observations?

Not necessarily. They are parameters of a probability distribution encoding prior belief.

Calling them “pseudo-counts” is an intuition for conjugate updating, not a claim that exactly $\alpha - 1$ real people or experiments were observed.

Thus:

pseudo-observation interpretation is a useful mathematical analogy, not a requirement.

7.2 Question 8.2 — Laplace's Rule of Succession

Exact question from the assignment:

Suppose

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta),$$

with

$$\theta \sim \text{Beta}(1, 1).$$

A sequence of n observations contains z successes.

(a) Derive the posterior distribution of θ .

(b) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(c) Compare this quantity with the MLE

$$\hat{\theta}_{\text{MLE}} = \frac{z}{n}.$$

(d) Show algebraically that

$$\frac{z+1}{n+2} = \frac{n}{n+2} \frac{z}{n} + \frac{2}{n+2} \frac{1}{2}.$$

(e) Interpret the posterior mean as shrinkage toward $\frac{1}{2}$.

(f) Examine the cases

$$(z, n) = (0, 1), (1, 1), (1, 2), (50, 100).$$

(g) Explain why the difference between Bayesian prediction and the MLE becomes negligible for large samples.

Theory required before solving

Under the uniform prior

$$\theta \sim \text{Beta}(1, 1),$$

and Bernoulli observations with z successes in n trials,

$$\theta \mid y \sim \text{Beta}(z+1, n-z+1).$$

The posterior predictive probability equals the posterior mean:

$$P(Y_{n+1} = 1 \mid y) = \frac{z+1}{n+2}.$$

The MLE is

$$\hat{\theta}_{\text{MLE}} = \frac{z}{n}.$$

The Bayesian expression is a weighted average:

$$\frac{z+1}{n+2} = \frac{n}{n+2} \frac{z}{n} + \frac{2}{n+2} \frac{1}{2}.$$

Step-by-step solution

(a) Posterior.

Bayes' rule gives

$$\pi(\theta | y) \propto \theta^z (1 - \theta)^{n-z}.$$

This is the Beta kernel with parameters

$$z + 1, \quad n - z + 1.$$

Thus

$$\theta | y \sim \text{Beta}(z + 1, n - z + 1).$$

(b) Predictive probability.

Condition on θ :

$$P(Y_{n+1} = 1 | \theta, y) = \theta.$$

Average over the posterior:

$$P(Y_{n+1} = 1 | y) = E(\theta | y).$$

The Beta mean is

$$P(Y_{n+1} = 1 | y) = \frac{z + 1}{n + 2}.$$

(c) Compare with the MLE.

The MLE is

$$\hat{\theta}_{\text{MLE}} = \frac{z}{n}.$$

The Bayesian value is

$$\frac{z + 1}{n + 2},$$

which is generally different because the Bayesian analysis includes prior information.

(d) Show the weighted-average identity.

Start with

$$\frac{n}{n+2} \frac{z}{n} + \frac{2}{n+2} \frac{1}{2}.$$

Simplify:

$$= \frac{z}{n+2} + \frac{1}{n+2} = \boxed{\frac{z+1}{n+2}}.$$

(e) Shrinkage interpretation.

The prior mean under $U(0, 1)$ is $1/2$. Therefore the posterior mean is a compromise between

$$\frac{z}{n} \quad \text{and} \quad \frac{1}{2}.$$

For n large,

$$\frac{n}{n+2} \rightarrow 1, \quad \frac{2}{n+2} \rightarrow 0.$$

Hence the data dominate and the shrinkage disappears.

(f) Numerical cases.

For $(z, n) = (0, 1)$:

$$\frac{z+1}{n+2} = \frac{1}{3} \approx 0.3333, \quad \frac{z}{n} = 0.$$

For $(1, 1)$:

$$\frac{2}{3} \approx 0.6667, \quad \frac{1}{1} = 1.$$

For $(1, 2)$:

$$\frac{3}{4} = 0.75, \quad \frac{1}{2} = 0.5.$$

For $(50, 100)$:

$$\frac{51}{102} = 0.5, \quad \frac{50}{100} = 0.5.$$

The last case illustrates how the prior pull can be tiny or even exactly cancel the data displacement from $1/2$.

(g) Large-sample difference.

Subtract:

$$\frac{z+1}{n+2} - \frac{z}{n} = \frac{n(z+1) - z(n+2)}{n(n+2)} = \frac{n-2z}{n(n+2)}.$$

The numerator is at most order n , while the denominator is order n^2 . Hence the difference is order $1/n$:

$$\boxed{\frac{z+1}{n+2} - \frac{z}{n} = O(1/n) \rightarrow 0.}$$

Thus Bayesian prediction and MLE prediction become asymptotically equivalent under fixed prior parameters.

7.3 Question 8.3 — Prior Predictive Distribution

Exact question from the assignment:

Suppose

$$\theta \sim U(0, 1),$$

and conditional on θ ,

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

Let

$$Z = \sum_{i=1}^n Y_i.$$

(a) Derive

$$P(Z = z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta.$$

(b) Evaluate the integral using the Beta function.

(c) Show that

$$P(Z = z) = \frac{1}{n+1}, \quad z = 0, \dots, n.$$

(d) Explain why this result may initially seem surprising.

(e) Compare the prior predictive distribution under

$$\theta \sim U(0, 1)$$

with the sampling distribution under a fixed

$$\theta = \frac{1}{2}.$$

(f) Explain the conceptual difference between uncertainty about θ and randomness conditional on θ .

Theory required before solving

The prior predictive distribution integrates out parameter uncertainty:

$$P(Z = z) = \int P(Z = z \mid \theta) \pi(\theta) d\theta.$$

If

$$Z \mid \theta \sim \text{Binomial}(n, \theta)$$

and $\theta \sim U(0, 1) = \text{Beta}(1, 1)$, then

$$P(Z = z) = \binom{n}{z} B(z + 1, n - z + 1).$$

The Beta function satisfies

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)}.$$

Step-by-step solution

(a) Derive the requested integral.

Conditional on θ ,

$$P(Z = z \mid \theta) = \binom{n}{z} \theta^z (1 - \theta)^{n-z}.$$

Because the prior is uniform,

$$\pi(\theta) = 1, \quad 0 < \theta < 1.$$

Therefore

$$P(Z = z) = \binom{n}{z} \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta.$$

(b) Use the Beta function.

By definition,

$$B(z + 1, n - z + 1) = \int_0^1 \theta^z (1 - \theta)^{n-z} d\theta.$$

Hence

$$P(Z = z) = \binom{n}{z} B(z + 1, n - z + 1).$$

Using factorial identities,

$$B(z + 1, n - z + 1) = \frac{z!(n - z)!}{(n + 1)!}.$$

Therefore

$$P(Z = z) = \frac{n!}{z!(n - z)!} \frac{z!(n - z)!}{(n + 1)!}.$$

(c) Simplify.

Cancel the factorials:

$$P(Z = z) = \frac{n!}{(n + 1)!} = \boxed{\frac{1}{n + 1}}, \quad z = 0, \dots, n.$$

Thus every possible success count is equally likely.

(d) Why surprising?

A uniform prior on θ is uniform over possible probabilities, not obviously uniform over

counts. Yet after integrating over θ , the combinatorial factor in the Binomial distribution and the Beta integral cancel exactly.

So the result is

$$Z \sim \text{Discrete Uniform}\{0, 1, \dots, n\}.$$

(e) Compare with fixed $\theta = 1/2$.

If $\theta = 1/2$ is fixed, then

$$Z \sim \text{Binomial}\left(n, \frac{1}{2}\right),$$

so

$$P(Z = z) = \binom{n}{z} 2^{-n}.$$

This distribution is concentrated around $n/2$.

Under the uniform prior on θ , however,

$$P(Z = z) = \frac{1}{n+1}$$

for every z . Extreme counts such as 0 and n are therefore much more plausible than under the fixed- $1/2$ model.

(f) Conceptual distinction.

With fixed θ , randomness means sampling variation conditional on a known parameter:

$$Z \mid \theta.$$

With Bayesian prior uncertainty, θ itself is random before observing the data. Therefore the prior predictive distribution combines two layers:

$$\text{uncertainty about } \theta + \text{sampling randomness given } \theta.$$

This is why the prior predictive can be much more dispersed than a sampling distribution at one fixed value of θ .

7.4 Question 9.1 — Posterior Mean and Variance

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta),$$

and

$$\theta \sim \text{Beta}(\alpha, \beta).$$

Let

$$Z = \sum_{i=1}^n Y_i.$$

- (a) Derive the posterior distribution.
- (b) Derive the posterior mean.
- (c) Show that the posterior mean can be written as a weighted average of the prior mean and the sample proportion.
- (d) Derive the posterior variance.
- (e) Assume

$$\frac{Z}{n} \rightarrow \theta_0.$$

Show heuristically that the posterior variance approaches zero as $n \rightarrow \infty$.

- (f) Explain why posterior concentration does not necessarily mean that every prior leads to the same posterior for finite n .

Theory required before solving

Beta-Bernoulli conjugacy gives

$$\theta \mid y \sim \text{Beta}(\alpha + z, \beta + n - z).$$

For $X \sim \text{Beta}(a, b)$,

$$E(X) = \frac{a}{a+b}, \quad \text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

The prior mean is $\theta_0 = \alpha/(\alpha + \beta)$, and the sample proportion is $\bar{Y} = z/n$.

Step-by-step solution

Let

$$z = \sum_{i=1}^n Y_i.$$

(a) Posterior distribution.

$$\pi(\theta | y) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1} \theta^z (1-\theta)^{n-z} = \theta^{\alpha+z-1} (1-\theta)^{\beta+n-z-1}.$$

Hence

$$\theta | y \sim \text{Beta}(\alpha + z, \beta + n - z).$$

(b) Posterior mean.

$$E(\theta | y) = \frac{\alpha + z}{\alpha + \beta + n}.$$

(c) Weighted-average form.

Since

$$\alpha = (\alpha + \beta) \frac{\alpha}{\alpha + \beta} = (\alpha + \beta) \theta_0$$

and

$$z = n\bar{Y},$$

we get

$$E(\theta | y) = \frac{(\alpha + \beta)\theta_0 + n\bar{Y}}{\alpha + \beta + n}.$$

Therefore

$$E(\theta | y) = \frac{\alpha + \beta}{\alpha + \beta + n} \theta_0 + \frac{n}{\alpha + \beta + n} \bar{Y}.$$

(d) Posterior variance.

Substitute

$$a = \alpha + z, \quad b = \beta + n - z$$

into the Beta variance:

$$\text{Var}(\theta | y) = \frac{(\alpha + z)(\beta + n - z)}{(\alpha + \beta + n)^2 (\alpha + \beta + n + 1)}.$$

(e) Why the variance tends to zero.

If $z/n \rightarrow \theta_0^{\text{true}}$, then for large n ,

$$\alpha + z \approx n \frac{z}{n}, \quad \beta + n - z \approx n \left(1 - \frac{z}{n}\right),$$

while the denominator is of order n^3 . Hence

$$\text{Var}(\theta | y) \approx \frac{\bar{Y}(1 - \bar{Y})}{n} \leq \frac{1}{4n} \rightarrow 0.$$

Thus

$$\text{Var}(\theta \mid y) \rightarrow 0.$$

(f) Why finite-sample priors can still differ.

For fixed n , the posterior mean explicitly contains

$$\frac{\alpha + \beta}{\alpha + \beta + n},$$

which is nonzero. Therefore two priors with different means or concentrations can give different posterior distributions for the same finite dataset.

Concentration is an asymptotic statement:

as $n \rightarrow \infty$ prior influence may vanish, but it need not vanish at finite n .

7.5 Question 9.2 — Extreme Priors

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_{10} \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta),$$

and

$$Z = 9.$$

Compare the following priors:

$$\theta \sim \text{Beta}(1, 1),$$

$$\theta \sim \text{Beta}(1000, 1000),$$

$$\theta \sim \text{Beta}(1000, 1).$$

- Derive all three posterior distributions.
- Compute or approximate the posterior means.
- Explain why the second prior may be considered highly concentrated around $\frac{1}{2}$.
- Explain why the third prior can dominate the observed data.
- For each prior, discuss what happens as the sample size becomes increasingly large while the sample proportion remains near 0.9.
- Explain the difference between a proper but highly informative prior and an improper prior.

Theory required before solving

Beta-Bernoulli conjugacy:

$$\text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + z, \beta + n - z).$$

The posterior mean is

$$\frac{\alpha + z}{\alpha + \beta + n}.$$

The Beta concentration $\alpha + \beta$ controls how strongly the prior pulls the posterior, while the prior mean is $\alpha/(\alpha + \beta)$. A proper prior integrates to one; an improper prior does not.

Step-by-step solution

Here

$$n = 10, \quad z = 9.$$

(a) Posteriors.

$$\text{Beta}(1, 1) \rightarrow \boxed{\text{Beta}(10, 2)}.$$

$$\text{Beta}(1000, 1000) \rightarrow \boxed{\text{Beta}(1009, 1001)}.$$

$$\text{Beta}(1000, 1) \rightarrow \boxed{\text{Beta}(1009, 2)}.$$

(b) Posterior means.

First:

$$E(\theta | y) = \frac{10}{12} = \boxed{0.8333}.$$

Second:

$$\frac{1009}{2010} \approx \boxed{0.5020}.$$

Third:

$$\frac{1009}{1011} \approx \boxed{0.9980}.$$

(c) Why Beta(1000, 1000) is concentrated near 1/2.

Its mean is

$$\frac{1000}{2000} = \frac{1}{2}.$$

Its variance is

$$\frac{1000^2}{2000^2 \cdot 2001} = \frac{1}{8004},$$

which is very small. Hence almost all prior probability lies close to 1/2.

(d) Why Beta(1000, 1) dominates.

Its mean is

$$\frac{1000}{1001} \approx 0.999,$$

and its concentration is 1001, vastly larger than the observed sample size 10. Thus the posterior mean remains near 1 even though the sample proportion is 0.9.

(e) Large-sample behavior.

For fixed α, β ,

$$E(\theta | y) = \frac{\alpha + \beta}{\alpha + \beta + n} \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \frac{z}{n}.$$

As $n \rightarrow \infty$,

$$\frac{n}{\alpha + \beta + n} \rightarrow 1.$$

If $z/n \rightarrow 0.9$, then

$$E(\theta | y) \rightarrow 0.9.$$

Also

$$\text{Var}(\theta | y) \rightarrow 0.$$

So each fixed proper prior is eventually overwhelmed by enough data under regular conditions.

(f) Proper informative versus improper.

A highly informative proper prior is a genuine probability distribution and can strongly affect finite-sample inference. An improper prior is not normalizable and is used only through proportionality. It is acceptable only when the posterior is proper:

$$\int \pi(\theta)L(\theta) \, d\theta < \infty.$$

Therefore

strong prior concentration and impropriety are fundamentally different ideas.

7.6 Question 9.3 — Exponential Family and Conjugacy

Exact question from the assignment:

The Binomial likelihood is

$$P(X = x | p) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

- (a) Rewrite the likelihood in exponential-family form.
- (b) Identify a natural parameter.
- (c) Identify a sufficient statistic.
- (d) Starting from a general exponential-family conjugate prior form, show why a Beta prior is conjugate.
- (e) Derive the posterior hyperparameter update.
- (f) Explain why conjugacy is computationally convenient.
- (g) Discuss one limitation of restricting Bayesian modeling only to conjugate families.

Theory required before solving

An exponential-family likelihood can be written as

$$f(x | \eta) = h(x) \exp\{\eta T(x) - A(\eta)\}.$$

For the Binomial model,

$$P(X = x | p) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

A natural parameter is the log-odds

$$\eta = \log \frac{p}{1 - p},$$

and the sufficient statistic is $T(X) = X$. In the original p -parameterization, the Beta family is conjugate because likelihood and prior are powers of p and $1 - p$.

Step-by-step solution

(a) Exponential-family form.

Start with

$$P(X = x | p) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

Write

$$x \log p + (n - x) \log(1 - p) = x \log \frac{p}{1 - p} + n \log(1 - p).$$

Define

$$\eta = \log \frac{p}{1 - p}.$$

Then

$$p = \frac{e^\eta}{1 + e^\eta}, \quad 1 - p = \frac{1}{1 + e^\eta}.$$

Therefore

$$P(X = x \mid \eta) = \binom{n}{x} \exp \{x\eta - n \log(1 + e^\eta)\}.$$

Thus

$$P(X = x \mid \eta) = h(x) \exp\{x\eta - A(\eta)\},$$

with $h(x) = \binom{n}{x}$ and $A(\eta) = n \log(1 + e^\eta)$.

(b) Natural parameter.

$$\eta = \log \frac{p}{1 - p}.$$

(c) Sufficient statistic.

The parameter-dependent data term is $x\eta$, so

$$T(X) = X.$$

For iid data, the sufficient statistic becomes $\sum_i X_i$.

(d) Why Beta is conjugate.

Take

$$\pi(p) \propto p^{\alpha-1} (1 - p)^{\beta-1}.$$

Multiplying by the likelihood,

$$\pi(p \mid x) \propto p^{\alpha-1+x} (1 - p)^{\beta-1+n-x}.$$

The result has exactly the same Beta form.

(e) Hyperparameter update.

Match exponents:

$$\alpha' - 1 = \alpha + x - 1, \quad \beta' - 1 = \beta + n - x - 1.$$

Hence

$$\alpha' = \alpha + x, \quad \beta' = \beta + n - x.$$

(f) Computational convenience.

Conjugacy lets us recognize the posterior family without evaluating a difficult normalizing integral. Consequently posterior means, variances and predictive probabilities follow from

known formulas.

(g) Limitation.

A convenient conjugate form can be scientifically unrealistic. Restricting all priors to conjugate families may exclude useful hierarchical, skewed, heavy-tailed, or structurally informed models.

Thus

conjugacy is a computational tool, not a criterion for choosing the scientific model.

Chapter 8

Normal–Normal Conjugacy and Prediction

8.1 Question 10.1 — Precision-Weighted Updating

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

where

$$\sigma^2 = 4.$$

The prior is

$$\theta \sim N(10, 9).$$

For

$$n = 25, \quad \bar{Y} = 12,$$

answer the following.

- (a) Derive the posterior variance.
- (b) Derive the posterior mean.
- (c) Express the posterior mean as

$$w\mu_0 + (1 - w)\bar{Y}.$$

- (d) Compute w .
- (e) Explain why the observation summary $\bar{Y}$ receives more weight than the prior mean.
- (f) Determine what happens to the posterior mean and variance as

$$n \rightarrow \infty.$$

(g) Determine what happens if

$$\tau_0^2 \rightarrow 0.$$

For

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2), \quad \theta \sim N(\mu_0, \tau_0^2),$$

the Normal–Normal conjugacy result is

$$\theta \mid y_1, \dots, y_n \sim N(\mu_n, \tau_n^2),$$

where

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

and

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}.$$

The quantities $1/\tau_0^2$ and n/σ^2 are respectively the prior precision and data precision. Thus precision adds, and the posterior mean is a precision-weighted average:

$$\mu_n = \frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2} \mu_0 + \frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2} \bar{y}.$$

This is the continuous analogue of Beta–Bernoulli shrinkage. As $n \rightarrow \infty$, the data precision dominates and

$$\mu_n \rightarrow \bar{y}, \quad \tau_n^2 \rightarrow 0.$$

We are given

$$\sigma^2 = 4, \quad \mu_0 = 10, \quad \tau_0^2 = 9, \quad n = 25, \quad \bar{y} = 12.$$

Step 1: Posterior variance.

Start from the precision formula:

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

Substituting the numerical values gives

$$\frac{1}{\tau_n^2} = \frac{1}{9} + \frac{25}{4}.$$

Use a common denominator:

$$\frac{1}{9} + \frac{25}{4} = \frac{4 + 225}{36} = \frac{229}{36}.$$

Therefore

$$\tau_n^2 = \frac{36}{229}.$$

Step 2: Posterior mean.

Use

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}.$$

Substitute:

$$\mu_n = \frac{10/9 + (25/4)(12)}{1/9 + 25/4}.$$

Because

$$\frac{25}{4}(12) = 75,$$

we obtain

$$\mu_n = \frac{10/9 + 75}{229/36} = \frac{685/9}{229/36}.$$

Multiplying numerator and denominator by 36,

$$\mu_n = \frac{2740}{229}.$$

Hence

$$\mu_n = \frac{2740}{229} \approx 11.9651.$$

The posterior mean is therefore very close to 12, which makes sense because 25 observations with variance 4 provide substantially more information than the prior with variance 9.

Step 3: Write the posterior mean as a weighted average.

Define

$$w = \frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2}.$$

Then the data weight is $1 - w$, and

$$\mu_n = w\mu_0 + (1 - w)\bar{y}.$$

Thus

$$\mu_n = w\mu_0 + (1 - w)\bar{y}.$$

Step 4: Compute w .

We have

$$w = \frac{1/9}{1/9 + 25/4} = \frac{1/9}{229/36} = \frac{4}{229}.$$

Therefore

$$w = \frac{4}{229} \approx 0.01747.$$

Consequently

$$1 - w = \frac{225}{229} \approx 0.98253.$$

Indeed,

$$\mu_n = \frac{4}{229}(10) + \frac{225}{229}(12) = \frac{40 + 2700}{229} = \frac{2740}{229},$$

which agrees with the direct calculation.

Step 5: Why does $\bar{Y}$ receive more weight?

The prior contributes precision

$$\frac{1}{9},$$

whereas the 25 observations contribute total precision

$$\frac{25}{4} = 6.25.$$

Since

$$6.25 \gg \frac{1}{9},$$

the data are much more precise than the prior information. Therefore the posterior mean must lie much closer to 12 than to 10.

This is not an arbitrary weighting rule: it comes directly from the reciprocals of the variances. Smaller variance means larger precision and therefore greater weight in the posterior.

Step 6: Behavior as $n \rightarrow \infty$.

For the variance,

$$\tau_n^2 = \frac{1}{1/\tau_0^2 + n/\sigma^2}.$$

As $n \rightarrow \infty$,

$$\frac{n}{\sigma^2} \rightarrow \infty,$$

so

$$\boxed{\tau_n^2 \rightarrow 0}.$$

For the mean,

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}.$$

The term involving $n\bar{y}$ eventually dominates the fixed prior term, so

$$\mu_n \rightarrow \bar{y}.$$

If the sample mean converges to the true parameter θ_0 , then also

$$\boxed{\mu_n \rightarrow \theta_0}.$$

Step 7: Behavior as $\tau_0^2 \rightarrow 0$.

If

$$\tau_0^2 \rightarrow 0,$$

then

$$\frac{1}{\tau_0^2} \rightarrow \infty.$$

Thus the prior precision dominates the data precision. Therefore

$$w = \frac{1/\tau_0^2}{1/\tau_0^2 + n/\sigma^2} \rightarrow 1,$$

and hence

$$\boxed{\mu_n \rightarrow \mu_0}, \quad \boxed{\tau_n^2 \rightarrow 0}.$$

So an infinitely concentrated prior effectively forces the posterior to remain at the prior location, because the model is treating the prior information as infinitely precise.

Interpretation

This problem is the clearest numerical illustration of precision-weighted Bayesian updating:

$$\boxed{\text{posterior precision} = \text{prior precision} + \text{data precision}.}$$

The posterior mean lies between the prior mean 10 and the observed mean 12, but it is much closer to 12 because the data carry much more precision.

8.2 Question 10.2 — Sequential Normal Updating

Exact question from the assignment:

Suppose

$$\theta \sim N(\mu_0, \tau_0^2),$$

and observations arrive one at a time:

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2).$$

- (a) Derive the posterior after observing Y_1 .
- (b) Treat this posterior as the prior for observing Y_2 and derive the next posterior.
- (c) Continue conceptually until n observations have been observed.
- (d) Show that the resulting posterior precision is

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

- (e) Show that sequential updating gives the same posterior as analyzing all observations simultaneously.
- (f) Explain why Bayesian updating can be performed online without storing all previous raw observations when sufficient statistics are available.

Bayesian updating is associative: the posterior obtained after one observation can be used as the prior for the next observation. For the Normal–Normal model,

$$\theta \sim N(\mu_0, \tau_0^2), \quad Y_i \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

the posterior after n observations is

$$\theta \mid y_{1:n} \sim N(\mu_n, \tau_n^2),$$

with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

and

$$\mu_n = \frac{\mu_0/\tau_0^2 + (1/\sigma^2) \sum_{i=1}^n y_i}{1/\tau_0^2 + n/\sigma^2}.$$

Thus each observation adds $1/\sigma^2$ to the precision. Because the likelihood factors,

$$p(y_{1:n} \mid \theta) = \prod_{i=1}^n p(y_i \mid \theta),$$

the order in which the data arrive does not change the final posterior.

Step 1: Posterior after Y_1 .

Start with

$$\theta \sim N(\mu_0, \tau_0^2).$$

After observing $Y_1 = y_1$, conjugacy gives

$$\theta \mid y_1 \sim N(\mu_1, \tau_1^2)$$

where

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2}$$

and

$$\mu_1 = \frac{\mu_0/\tau_0^2 + y_1/\sigma^2}{1/\tau_0^2 + 1/\sigma^2}.$$

The important idea is that the first observation simply contributes one additional unit of data precision.

Step 2: Use the first posterior as the prior for Y_2 .

After observing Y_1 , the new prior is

$$\theta \mid y_1 \sim N(\mu_1, \tau_1^2).$$

Now observe

$$Y_2 = y_2.$$

Applying the same conjugacy theorem again,

$$\theta \mid y_1, y_2 \sim N(\mu_2, \tau_2^2),$$

with

$$\frac{1}{\tau_2^2} = \frac{1}{\tau_1^2} + \frac{1}{\sigma^2}.$$

Substitute the expression for $1/\tau_1^2$:

$$\frac{1}{\tau_2^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2} + \frac{1}{\sigma^2} = \frac{1}{\tau_0^2} + \frac{2}{\sigma^2}.$$

For the mean,

$$\mu_2 = \frac{\mu_1/\tau_1^2 + y_2/\sigma^2}{1/\tau_1^2 + 1/\sigma^2}.$$

But

$$\frac{\mu_1}{\tau_1^2} = \frac{\mu_0}{\tau_0^2} + \frac{y_1}{\sigma^2},$$

so

$$\mu_2 = \frac{\mu_0/\tau_0^2 + y_1/\sigma^2 + y_2/\sigma^2}{1/\tau_0^2 + 2/\sigma^2}.$$

Step 3: Continue to n observations.

Every new observation adds another $1/\sigma^2$ to the precision and another y_i/σ^2 to the precision-weighted numerator.

Therefore,

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

and

$$\mu_n = \frac{\mu_0/\tau_0^2 + \sum_{i=1}^n y_i/\sigma^2}{1/\tau_0^2 + n/\sigma^2}.$$

Since

$$\sum_{i=1}^n y_i = n\bar{y},$$

this becomes

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}.$$

Step 4: Posterior precision.

From the sequential argument above,

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{1}{\sigma^2},$$

$$\frac{1}{\tau_2^2} = \frac{1}{\tau_1^2} + \frac{1}{\sigma^2},$$

and by induction,

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}.$$

This is the exact statement that information accumulates additively.

Step 5: Why is the sequential posterior identical to the simultaneous posterior?

The simultaneous posterior is

$$p(\theta \mid y_1, \dots, y_n) \propto p(\theta) \prod_{i=1}^n p(y_i \mid \theta).$$

The sequential calculation gives

$$p(\theta \mid y_1) \propto p(\theta)p(y_1 \mid \theta),$$

then

$$p(\theta \mid y_1, y_2) \propto p(\theta \mid y_1)p(y_2 \mid \theta) \propto p(\theta)p(y_1 \mid \theta)p(y_2 \mid \theta),$$

and continuing,

$$p(\theta \mid y_{1:n}) \propto p(\theta) \prod_{i=1}^n p(y_i \mid \theta).$$

This is exactly the same expression as the simultaneous analysis. Therefore

$$\text{sequential updating} = \text{batch updating}.$$

Step 6: Why can we perform online Bayesian updating?

The posterior depends on the observations only through

$$\sum_{i=1}^n y_i \quad \text{or equivalently} \quad \bar{y}.$$

Hence, instead of storing every previous observation, one may maintain the running sufficient statistic

$$S_n = \sum_{i=1}^n y_i.$$

When a new observation y_{n+1} arrives,

$$S_{n+1} = S_n + y_{n+1}.$$

Thus the posterior can be updated using only the current posterior parameters or sufficient statistics.

Final Insight

The central principle is

$$\text{Bayesian learning can be performed one observation at a time.}$$

This works because Bayes' rule is compatible with factorization of the likelihood. In a conjugate model, the update is especially simple because the posterior remains in the same family after every new observation.

8.3 Question 10.3 — Prior-Data Conflict

Exact question from the assignment:

Suppose

$$\theta \sim N(0, 1),$$

and

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, 1).$$

You observe

$$n = 5, \quad \bar{Y} = 10.$$

- Derive the posterior distribution.
- Compute the posterior mean.
- Explain why the posterior mean is neither 0 nor 10.
- Now suppose the prior variance changes from 1 to 100. Derive the new posterior.
- Compare the two posterior means.
- Explain how increasing prior variance represents weaker prior certainty.
- Discuss whether a prior that is strongly contradicted by the data should automatically be discarded.

For Normal–Normal conjugacy,

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2), \quad \theta \sim N(\mu_0, \tau_0^2),$$

the posterior is

$$\theta \mid y_{1:n} \sim N(\mu_n, \tau_n^2),$$

with

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2},$$

$$\mu_n = \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}.$$

Thus the posterior is a compromise between prior information and observed data. The relative influence of the two sources is determined by their precision, not simply by their numerical location.

A larger prior variance means weaker prior certainty. A smaller prior variance means stronger prior certainty.

Step 1: Posterior under $\theta \sim N(0, 1)$.

We have

$$\mu_0 = 0, \quad \tau_0^2 = 1, \quad \sigma^2 = 1, \quad n = 5, \quad \bar{y} = 10.$$

Hence

$$\frac{1}{\tau_n^2} = 1 + \frac{5}{1} = 6,$$

so

$$\tau_n^2 = \frac{1}{6}.$$

For the mean,

$$\mu_n = \frac{0/1 + (5/1)(10)}{1 + 5} = \frac{50}{6} = \frac{25}{3}.$$

Therefore

$$\theta \mid Y \sim N\left(\frac{25}{3}, \frac{1}{6}\right).$$

Step 2: Posterior mean.

From the calculation above,

$$E[\theta \mid Y] = \frac{25}{3} \approx 8.333.$$

Step 3: Why is the posterior mean neither 0 nor 10?

The prior mean is 0, while the data suggest 10. Neither source is infinitely precise.

The prior has precision

$$\frac{1}{\tau_0^2} = 1,$$

whereas the five observations have total precision

$$\frac{n}{\sigma^2} = 5.$$

Therefore the data get five times as much precision weight as the prior.

Indeed,

$$\mu_n = \frac{1}{6}(0) + \frac{5}{6}(10) = \frac{25}{3}.$$

Thus the posterior lies strictly between the two sources of information.

Step 4: Change the prior variance from 1 to 100.

Now

$$\mu_0 = 0, \quad \tau_0^2 = 100.$$

Hence

$$\frac{1}{\tau_n^2} = \frac{1}{100} + 5 = \frac{501}{100}.$$

Therefore

$$\tau_n^2 = \frac{100}{501}.$$

For the mean,

$$\mu_n = \frac{0/100 + 5(10)}{1/100 + 5}.$$

Thus

$$\mu_n = \frac{50}{501/100} = \frac{5000}{501} \approx 9.98004.$$

Hence

$$\theta \mid Y \sim N\left(\frac{5000}{501}, \frac{100}{501}\right).$$

Step 5: Compare the two posterior means.

Under the first prior,

$$\mu_n = \frac{25}{3} \approx 8.333.$$

Under the much more diffuse prior,

$$\mu_n = \frac{5000}{501} \approx 9.980.$$

Therefore

$$9.980 > 8.333.$$

The second posterior is much closer to the data because the prior no longer claims that values near 0 are especially plausible.

Step 6: Interpret increasing prior variance.

A variance of 1 means the prior density is concentrated around 0. A variance of 100 is much more spread out:

$$N(0, 100)$$

places far less relative mass near any small neighborhood of 0 than $N(0, 1)$.

Equivalently,

$$\frac{1}{1} = 1 \quad \text{versus} \quad \frac{1}{100} = 0.01.$$

Hence the prior precision falls by a factor of 100.

Therefore increasing τ_0^2 represents weaker prior certainty and gives the data greater relative influence.

Step 7: Should a contradicted prior automatically be discarded?

No. A Bayesian prior is part of the probability model and may encode genuine previous information. Strong disagreement between prior and data is a reason to investigate the assumptions, not automatically to throw away the prior.

A useful diagnostic is to ask whether the prior was based on valid historical information and whether the new data come from a comparable population.

In this example, replacing variance 1 by variance 100 is a principled way of saying:

“I still center my prior at 0, but I am much less certain about it.”

The posterior then allows the data to dominate without pretending that prior knowledge never existed.

Key Interpretation

Prior–data conflict is resolved through the same precision-weighting mechanism used everywhere in the Normal–Normal model:

more precise source $\implies$ greater posterior influence.

8.4 Question 11.1 — Marginal Distribution

Exact question from the assignment:

Suppose

$$\theta \sim N(\mu_0, \tau_0^2),$$

and

$$Y \mid \theta \sim N(\theta, \sigma^2).$$

(a) Write

$$Y = \theta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \sigma^2).$$

(b) Derive the marginal distribution of Y .

(c) Compute

$$E(Y).$$

(d) Compute

$$\text{Var}(Y)$$

using the law of total variance.

(e) Show that

$$\text{Var}(Y) = \tau_0^2 + \sigma^2.$$

(f) Interpret the two components of the marginal variance.

The hierarchical Normal model is

$$\theta \sim N(\mu_0, \tau_0^2), \quad Y \mid \theta \sim N(\theta, \sigma^2).$$

It can be written as

$$Y = \theta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2), \quad \theta \perp \varepsilon.$$

The sum of independent Normal random variables is Normal, with means and variances adding:

$$U \sim N(m_1, s_1^2), \quad V \sim N(m_2, s_2^2), \quad U \perp V$$

implies

$$U + V \sim N(m_1 + m_2, s_1^2 + s_2^2).$$

Also, the law of total variance states

$$\text{Var}(Y) = \text{E}[\text{Var}(Y \mid \theta)] + \text{Var}(\text{E}[Y \mid \theta]).$$

Step 1: Write Y as signal plus noise.

From

$$Y \mid \theta \sim N(\theta, \sigma^2),$$

we may write

$$Y = \theta + \varepsilon,$$

where

$$\varepsilon \sim N(0, \sigma^2)$$

and ε is independent of θ .

Step 2: Derive the marginal distribution of Y .

We have

$$\theta \sim N(\mu_0, \tau_0^2), \quad \varepsilon \sim N(0, \sigma^2), \quad \theta \perp \varepsilon.$$

Therefore

$$Y = \theta + \varepsilon$$

is Normal with

$$\text{E}[Y] = \text{E}[\theta] + \text{E}[\varepsilon] = \mu_0 + 0 = \mu_0$$

and

$$\text{Var}(Y) = \text{Var}(\theta) + \text{Var}(\varepsilon) = \tau_0^2 + \sigma^2.$$

Hence

$$Y \sim N(\mu_0, \tau_0^2 + \sigma^2).$$

Step 3: Compute $\text{E}(Y)$.

Using linearity of expectation,

$$\text{E}(Y) = \text{E}(\theta + \varepsilon) = \text{E}(\theta) + \text{E}(\varepsilon) = \mu_0.$$

Thus

$$\text{E}(Y) = \mu_0.$$

Step 4: Compute $\text{Var}(Y)$ using total variance.

First,

$$\text{E}[Y \mid \theta] = \theta, \quad \text{Var}(Y \mid \theta) = \sigma^2.$$

Therefore

$$\text{Var}(Y) = \text{E}[\text{Var}(Y \mid \theta)] + \text{Var}(\text{E}[Y \mid \theta]).$$

Substitute:

$$\text{Var}(Y) = \text{E}[\sigma^2] + \text{Var}(\theta) = \sigma^2 + \tau_0^2.$$

Hence

$$\boxed{\text{Var}(Y) = \tau_0^2 + \sigma^2}.$$

Step 5: Interpret the two components.

The marginal variance

$$\tau_0^2 + \sigma^2$$

contains two conceptually different sources of uncertainty.

The term

$$\boxed{\tau_0^2}$$

is uncertainty about the latent parameter θ before observing Y .

The term

$$\boxed{\sigma^2}$$

is irreducible observational noise: even if θ were known exactly, a future observation would still fluctuate around θ .

Thus

$$\boxed{\text{total predictive uncertainty} = \text{parameter uncertainty} + \text{observation noise.}}$$

8.5 Question 11.2 — Dependence in a Hierarchical Normal Model

Exact question from the assignment:

Suppose

$$Y_i = \theta + \varepsilon_i,$$

where

$$\theta \sim N(\mu_0, \tau_0^2),$$

and

$$\varepsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2).$$

(a) Compute

$$E(Y_i).$$

(b) Compute

$$\text{Var}(Y_i).$$

(c) Compute

$$\text{Cov}(Y_i, Y_j), \quad i \neq j.$$

(d) Compute the correlation between two observations.

(e) Explain why the observations are independent conditional on θ but correlated marginally.

(f) Discuss how this normal example is analogous to the Beta–Bernoulli model.

Consider

$$Y_i = \theta + \varepsilon_i,$$

where

$$\theta \sim N(\mu_0, \tau_0^2), \quad \varepsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2),$$

and the noise variables are independent of θ .

Conditional on θ ,

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

so the observations are conditionally independent.

Marginally, however, they share the common random component θ . A standard covariance identity gives

$$\text{Cov}(Y_i, Y_j) = \text{Cov}(E[Y_i \mid \theta], E[Y_j \mid \theta]) + E[\text{Cov}(Y_i, Y_j \mid \theta)].$$

For $i \neq j$, the second term is zero, while the first is $\text{Var}(\theta)$. Therefore

$$\text{Cov}(Y_i, Y_j) = \tau_0^2.$$

Step 1: Compute $E(Y_i)$.

Since

$$Y_i = \theta + \varepsilon_i,$$

we have

$$E(Y_i) = E(\theta) + E(\varepsilon_i) = \mu_0.$$

Thus

$$\boxed{E(Y_i) = \mu_0}.$$

Step 2: Compute $\text{Var}(Y_i)$.

Because θ and ε_i are independent,

$$\text{Var}(Y_i) = \text{Var}(\theta) + \text{Var}(\varepsilon_i) = \tau_0^2 + \sigma^2.$$

Hence

$$\boxed{\text{Var}(Y_i) = \tau_0^2 + \sigma^2}.$$

Step 3: Compute $\text{Cov}(Y_i, Y_j)$ for $i \neq j$.

Write

$$Y_i = \theta + \varepsilon_i, \quad Y_j = \theta + \varepsilon_j.$$

Then

$$\text{Cov}(Y_i, Y_j) = \text{Cov}(\theta + \varepsilon_i, \theta + \varepsilon_j).$$

Expanding bilinearly,

$$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta) + \text{Cov}(\theta, \varepsilon_j) + \text{Cov}(\varepsilon_i, \theta) + \text{Cov}(\varepsilon_i, \varepsilon_j).$$

All cross terms are zero by independence, including

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$$

for $i \neq j$. Therefore

$$\boxed{\text{Cov}(Y_i, Y_j) = \tau_0^2}.$$

Step 4: Compute the correlation.

By definition,

$$\text{Corr}(Y_i, Y_j) = \frac{\text{Cov}(Y_i, Y_j)}{\sqrt{\text{Var}(Y_i) \text{Var}(Y_j)}}.$$

Both variances are $\tau_0^2 + \sigma^2$, so

$$\text{Corr}(Y_i, Y_j) = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}.$$

Hence

$$\boxed{\text{Corr}(Y_i, Y_j) = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}.$$

Step 5: Conditional independence versus marginal dependence.

Given θ ,

$$Y_i \perp Y_j \mid \theta$$

because the only randomness remaining comes from independent errors.

Before observing θ , both variables contain the same random component θ . If θ happens to be large, both observations tend to be large; if θ is small, both tend to be small.

Therefore they are generally dependent marginally:

$$\boxed{Y_i \perp Y_j \mid \theta \quad \text{but} \quad Y_i \not\perp Y_j}.$$

Step 6: Analogy with Beta–Bernoulli.

In the Beta–Bernoulli model,

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta),$$

and the common latent parameter produces marginal dependence:

$$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta).$$

The Normal model has exactly the same structural idea:

$$Y_i = \theta + \varepsilon_i, \quad \text{Cov}(Y_i, Y_j) = \text{Var}(\theta).$$

Thus the recurring hierarchical pattern is

$$\boxed{\text{conditional independence} + \text{shared latent parameter} \implies \text{marginal dependence.}}$$

8.6 Question 11.3 — Posterior Predictive Distribution

Exact question from the assignment:

Suppose after observing data Y you obtain

$$\theta \mid Y \sim N(\mu_n, \tau_n^2).$$

A future observation satisfies

$$\tilde{Y} \mid \theta \sim N(\theta, \sigma^2).$$

(a) Derive

$$E(\tilde{Y} \mid Y).$$

(b) Derive

$$\text{Var}(\tilde{Y} \mid Y).$$

(c) Hence identify the posterior predictive distribution.

(d) Explain why

$$\text{Var}(\tilde{Y} \mid Y) > \text{Var}(\theta \mid Y).$$

(e) Suppose

$$\tau_n^2 \rightarrow 0.$$

What does the predictive distribution converge to?

(f) Explain the difference between estimating θ accurately and predicting an individual future observation accurately.

Suppose the data Y produce a posterior

$$\theta \mid Y \sim N(\mu_n, \tau_n^2),$$

and a future observation satisfies

$$\tilde{Y} \mid \theta \sim N(\theta, \sigma^2).$$

The posterior predictive distribution integrates over posterior uncertainty in θ :

$$p(\tilde{Y} \mid Y) = \int p(\tilde{Y} \mid \theta, Y) p(\theta \mid Y) d\theta.$$

Conditional independence gives

$$p(\tilde{Y} \mid \theta, Y) = p(\tilde{Y} \mid \theta).$$

Equivalently,

$$\tilde{Y} = \theta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2),$$

with ε independent of the posterior draw of θ . Hence the posterior predictive distribution is Normal with mean μ_n and variance $\tau_n^2 + \sigma^2$.

Step 1: Predictive mean.

Use total expectation:

$$\mathbb{E}(\tilde{Y} \mid Y) = \mathbb{E} \left[\mathbb{E}(\tilde{Y} \mid \theta, Y) \mid Y \right].$$

Because

$$\tilde{Y} \mid \theta, Y \sim N(\theta, \sigma^2),$$

we have

$$\mathbb{E}(\tilde{Y} \mid \theta, Y) = \theta.$$

Therefore

$$\mathbb{E}(\tilde{Y} \mid Y) = \mathbb{E}(\theta \mid Y) = \mu_n.$$

Thus

$$\boxed{\mathbb{E}(\tilde{Y} \mid Y) = \mu_n}.$$

Step 2: Predictive variance.

Use the law of total variance:

$$\text{Var}(\tilde{Y} \mid Y) = \mathbb{E}[\text{Var}(\tilde{Y} \mid \theta, Y) \mid Y] + \text{Var}[\mathbb{E}(\tilde{Y} \mid \theta, Y) \mid Y].$$

The first term is

$$\mathbb{E}[\sigma^2 \mid Y] = \sigma^2.$$

The second term is

$$\text{Var}(\theta \mid Y) = \tau_n^2.$$

Hence

$$\boxed{\text{Var}(\tilde{Y} \mid Y) = \sigma^2 + \tau_n^2}.$$

Step 3: Identify the predictive distribution.

The future observation is the sum of two independent Normal quantities:

$$\tilde{Y} = \theta + \varepsilon,$$

where, under the posterior,

$$\theta \mid Y \sim N(\mu_n, \tau_n^2), \quad \varepsilon \sim N(0, \sigma^2).$$

Therefore

$$\tilde{Y} \mid Y \sim N(\mu_n, \tau_n^2 + \sigma^2).$$

Step 4: Why is predictive variance larger than posterior variance?

We have

$$\text{Var}(\theta \mid Y) = \tau_n^2$$

but

$$\text{Var}(\tilde{Y} \mid Y) = \tau_n^2 + \sigma^2.$$

Since $\sigma^2 > 0$,

$$\text{Var}(\tilde{Y} \mid Y) > \text{Var}(\theta \mid Y).$$

The extra σ^2 is the new observation noise. Learning θ does not remove randomness from the measurement process itself.

Step 5: What happens if $\tau_n^2 \rightarrow 0$?

Then posterior uncertainty about θ disappears:

$$\tau_n^2 \rightarrow 0.$$

Therefore

$$\tilde{Y} \mid Y \Rightarrow N(\mu_n, \sigma^2).$$

Thus

$$\tilde{Y} \mid Y \rightarrow N(\mu_n, \sigma^2).$$

There is still variance σ^2 , because future observations remain noisy.

Step 6: Estimate versus predict.

Estimating θ asks:

“Where is the latent mean?”

As data accumulate, the posterior for θ can become extremely concentrated.

Predicting $\tilde{Y}$ asks:

“What actual noisy observation will I see next?”

Even with perfect knowledge of θ , a new observation still contains measurement noise.

Therefore

parameter uncertainty can vanish while predictive uncertainty remains.

Key Interpretation

The posterior predictive variance has a fundamental decomposition:

$$\underbrace{\tau_n^2}_{\text{uncertainty about } \theta} + \underbrace{\sigma^2}_{\text{irreducible observation noise}}.$$

This is one of the most important distinctions in Bayesian prediction.

Chapter 9

Objective Bayesian Analysis

9.1 Question 12.1 — When an Improper Prior Works

Exact question from the assignment:

Suppose

$$X_1, \dots, X_n \mid \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

where σ^2 is known.

Use the improper prior

$$\pi(\theta) \propto 1, \quad -\infty < \theta < \infty.$$

- (a) Show that the prior cannot be normalized.
- (b) Derive the posterior kernel.
- (c) Show that the posterior is proper.
- (d) Identify the posterior distribution.
- (e) Explain why

$$\pi(\theta) \propto 1$$

is sometimes interpreted as a limiting case of

$$\theta \sim N(0, \tau^2)$$

as

$$\tau^2 \rightarrow \infty.$$

- (f) Explain why the propriety of the posterior must be checked rather than assumed.

A prior $\pi(\theta)$ is called improper if

$$\pi(\theta) \geq 0 \quad \text{but} \quad \int \pi(\theta) d\theta = \infty.$$

The key Bayesian requirement is not that the prior itself integrate to one, but that the posterior can be normalized:

$$\pi(\theta | x) \propto \pi(\theta)f(x | \theta).$$

Thus the posterior is proper precisely when

$$\int \pi(\theta)f(x | \theta) d\theta < \infty.$$

For the Normal mean model,

$$X_1, \dots, X_n | \theta \stackrel{\text{ind}}{\sim} N(\theta, \sigma^2),$$

the improper flat prior

$$\pi(\theta) \propto 1$$

produces the proper posterior

$$\theta | x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

Step 1: Show the flat prior cannot be normalized.

We are given

$$\pi(\theta) \propto 1, \quad -\infty < \theta < \infty.$$

To normalize it, we would need a finite constant C such that

$$\int_{-\infty}^{\infty} C d\theta = 1.$$

But

$$\int_{-\infty}^{\infty} 1 d\theta = \infty.$$

Therefore no finite normalizing constant exists:

$\pi(\theta) \propto 1 \text{ is improper.}$

Step 2: Derive the posterior kernel.

The likelihood is

$$p(x_{1:n} | \theta) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\}.$$

Since the prior is constant,

$$\pi(\theta | x) \propto \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\}.$$

Use the identity

$$\sum_{i=1}^n (x_i - \theta)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \theta)^2.$$

The first term does not depend on θ , so it can be absorbed into the proportionality constant. Hence

$$\pi(\theta | x) \propto \exp \left\{ -\frac{n}{2\sigma^2} (\theta - \bar{x})^2 \right\}.$$

Step 3: Show the posterior is proper.

The kernel

$$\exp \left\{ -\frac{n}{2\sigma^2} (\theta - \bar{x})^2 \right\}$$

is the kernel of a Normal density with finite variance σ^2/n . Its integral over the real line is

$$\sqrt{\frac{2\pi\sigma^2}{n}} < \infty.$$

Therefore it can be normalized.

Thus

the posterior is proper.

Step 4: Identify the posterior distribution.

Matching with the Normal kernel gives

$$\theta | x_{1:n} \sim N \left(\bar{x}, \frac{\sigma^2}{n} \right).$$

Step 5: Explain the limiting Normal prior interpretation.

Start with the proper prior

$$\theta \sim N(0, \tau^2).$$

Its precision is $1/\tau^2$. The conjugate posterior has

$$\frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2},$$

and

$$\mu_n = \frac{(1/\tau^2)0 + (n/\sigma^2)\bar{x}}{1/\tau^2 + n/\sigma^2}.$$

As

$$\tau^2 \rightarrow \infty,$$

we have

$$\frac{1}{\tau^2} \rightarrow 0.$$

Therefore

$$\tau_n^2 \rightarrow \frac{\sigma^2}{n}, \quad \mu_n \rightarrow \bar{x}.$$

Hence the posterior converges exactly to

$$N\left(\bar{x}, \frac{\sigma^2}{n}\right).$$

Thus the flat improper prior can be viewed as the limiting case of an increasingly diffuse proper Normal prior.

Step 6: Why must propriety be checked?

The prior being improper means there is no probability distribution associated with it by itself. It is only a formal prior density shape.

Therefore the expression

$$\pi(\theta | x) \propto \pi(\theta)L(\theta)$$

is meaningful only if the resulting kernel is integrable. If

$$\int \pi(\theta)L(\theta) d\theta = \infty,$$

then no normalized posterior exists.

So the correct rule is

improper prior may be used, but posterior propriety must be verified.

9.2 Question 12.2 — When an Improper Prior Fails

Exact question from the assignment:

Suppose

$$X \mid \theta \sim U(0, \theta), \quad \theta > 0.$$

Consider the improper flat prior

$$\pi(\theta) \propto 1.$$

- (a) Write the likelihood.
- (b) Derive the posterior kernel.
- (c) Determine whether

$$\int_0^\infty \pi(\theta) p(x \mid \theta) d\theta$$

is finite.

- (d) Conclude whether the posterior is proper.
- (e) Now consider

$$\pi(\theta) \propto \frac{1}{\theta}.$$

- (f) Derive the corresponding posterior.
- (g) Determine whether it is proper.
- (h) Explain why “noninformative” does not mean that every improper prior is automatically acceptable.

If

$$X \mid \theta \sim U(0, \theta), \quad \theta > 0,$$

then

$$p(x \mid \theta) = \frac{1}{\theta} \mathbf{1}\{0 < x < \theta\}.$$

With an improper prior $\pi(\theta) \propto 1$,

$$\pi(\theta \mid x) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

This posterior is proper only if

$$\int_x^\infty \frac{1}{\theta} d\theta < \infty.$$

But the integral diverges.

Changing to

$$\pi(\theta) \propto \frac{1}{\theta}$$

changes the posterior kernel to

$$\theta^{-2} \mathbf{1}\{\theta > x\},$$

whose integral is finite for $x > 0$. This demonstrates that “improper” does not automatically mean “usable” or “noninformative.”

Step 1: Write the likelihood.

If

$$X \mid \theta \sim U(0, \theta),$$

then the density is

$$p(x \mid \theta) = \frac{1}{\theta} \mathbf{1}\{0 < x < \theta\}.$$

Since x has already been observed and is positive, the condition

$$0 < x < \theta$$

is equivalent to

$$\theta > x.$$

Therefore

$$p(x \mid \theta) = \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

Step 2: Posterior under the flat improper prior.

Take

$$\pi(\theta) \propto 1.$$

Then

$$\pi(\theta \mid x) \propto \pi(\theta)p(x \mid \theta) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

Thus the posterior kernel is

$$\pi(\theta \mid x) \propto \theta^{-1} \mathbf{1}\{\theta > x\}.$$

Step 3: Check integrability.

We must evaluate

$$\int_0^\infty \pi(\theta)p(x \mid \theta) d\theta.$$

Because the kernel vanishes below x ,

$$\int_0^\infty \frac{1}{\theta} \mathbf{1}\{\theta > x\} d\theta = \int_x^\infty \frac{1}{\theta} d\theta.$$

Now

$$\int_x^\infty \frac{1}{\theta} d\theta = [\log \theta]_x^\infty = \infty.$$

Therefore

the posterior kernel is not integrable.

Step 4: Conclusion.

Since the integral is infinite, no normalizing constant exists. Hence

the posterior is improper and unusable.

Step 5: Use $\pi(\theta) \propto 1/\theta$ instead.

Now

$$\pi(\theta) \propto \frac{1}{\theta}.$$

Bayes' rule gives

$$\pi(\theta | x) \propto \frac{1}{\theta} \cdot \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

Therefore

$$\pi(\theta | x) \propto \theta^{-2} \mathbf{1}\{\theta > x\}.$$

Step 6: Check propriety.

We need

$$\int_x^\infty \theta^{-2} d\theta.$$

Compute:

$$\int_x^\infty \theta^{-2} d\theta = \left[-\theta^{-1} \right]_x^\infty = \frac{1}{x} < \infty$$

for $x > 0$.

Therefore the posterior is proper.

Step 7: Normalize the posterior.

The kernel is

$$\theta^{-2} \mathbf{1}\{\theta > x\},$$

and its integral is $1/x$. Thus multiply by x :

$$\pi(\theta | x) = x\theta^{-2} \mathbf{1}\{\theta > x\}.$$

A direct verification is

$$\int_x^\infty x\theta^{-2} d\theta = x \cdot \frac{1}{x} = 1.$$

Step 8: Interpretation.

Both priors are improper:

$$1, \quad \frac{1}{\theta}.$$

But they lead to fundamentally different posteriors. Therefore the label “noninformative” cannot substitute for an actual propriety check.

The lesson is

improper prior $\nRightarrow$ improper posterior,

and also

improper prior $\nRightarrow$ acceptable posterior.

9.3 Question 12.3 — Diffuse Proper vs Improper

Exact question from the assignment:

Suppose

$$Y \mid \theta \sim N(\theta, 1).$$

Compare the priors

$$\theta \sim N(0, 10),$$

$$\theta \sim N(0, 10^6),$$

and

$$\pi(\theta) \propto 1.$$

- (a) Derive the posterior distribution under each proper prior.
- (b) Determine the limiting posterior as the prior variance tends to infinity.
- (c) Compare the limiting posterior with the posterior obtained using the improper flat prior.
- (d) Explain why a highly diffuse normal prior is not exactly the same mathematical object as an improper prior.
- (e) Discuss whether increasing the variance of a proper prior always makes it “less informative” under every possible reparameterization.

For the Normal–Normal model

$$Y \mid \theta \sim N(\theta, 1), \quad \theta \sim N(0, \tau^2),$$

the posterior is

$$\theta \mid Y = y \sim N\left(\frac{\tau^2}{\tau^2 + 1}y, \frac{\tau^2}{\tau^2 + 1}\right).$$

As

$$\tau^2 \rightarrow \infty,$$

the prior precision $1/\tau^2 \rightarrow 0$, so the posterior approaches

$$N(y, 1).$$

This is the posterior produced by the flat improper prior

$$\pi(\theta) \propto 1.$$

However, a diffuse proper prior is still a genuine probability distribution for every finite τ^2 , whereas the flat prior is not.

Step 1: Prior $\theta \sim N(0, 10)$.

Here

$$\tau^2 = 10.$$

Therefore

$$\mu_{\text{post}} = \frac{10}{10 + 1}y = \frac{10}{11}y,$$

and

$$\tau_{\text{post}}^2 = \frac{10}{11}.$$

Hence

$$\theta \mid y \sim N\left(\frac{10}{11}y, \frac{10}{11}\right).$$

Step 2: Prior $\theta \sim N(0, 10^6)$.

Now

$$\tau^2 = 10^6.$$

Thus

$$\mu_{\text{post}} = \frac{10^6}{10^6 + 1}y,$$

and

$$\tau_{\text{post}}^2 = \frac{10^6}{10^6 + 1}.$$

Therefore

$$\theta \mid y \sim N\left(\frac{10^6}{10^6 + 1}y, \frac{10^6}{10^6 + 1}\right).$$

Numerically,

$$\frac{10^6}{10^6 + 1} \approx 0.999999.$$

So the posterior is already extremely close to

$$N(y, 1).$$

Step 3: Flat improper prior.

With

$$\pi(\theta) \propto 1,$$

the likelihood is

$$p(y \mid \theta) \propto \exp\left\{-\frac{1}{2}(y - \theta)^2\right\}.$$

Therefore

$$\theta \mid y \sim N(y, 1).$$

Step 4: Determine the limiting posterior.

Take

$$\tau^2 \rightarrow \infty.$$

Then

$$\frac{\tau^2}{\tau^2 + 1} \rightarrow 1.$$

Hence

$$\mu_{\text{post}} \rightarrow y, \quad \tau_{\text{post}}^2 \rightarrow 1.$$

Thus

$$\theta \mid y \Rightarrow N(y, 1).$$

This matches the flat-prior posterior exactly.

Step 5: Why are the proper diffuse and improper priors not the same object?

For every finite τ^2 ,

$$N(0, \tau^2)$$

integrates to 1. It is a genuine probability distribution.

By contrast,

$$\pi(\theta) \propto 1$$

has

$$\int_{-\infty}^{\infty} 1 \, d\theta = \infty.$$

It is not a probability distribution at all.

Thus the flat prior is the mathematical limit of diffuse Normal priors, but it is not itself a member of that proper family.

Step 6: Does larger variance always mean less informative under every parameterization?

No. A prior that is diffuse in one coordinate need not be diffuse in a nonlinear transformation of that coordinate. Flatness and diffuseness are parameterization-dependent notions.

This is one of the reasons Jeffreys priors are introduced: they are constructed to transform coherently under smooth one-to-one reparameterizations.

Conclusion

The three analyses form a useful hierarchy:

$$N(0, 10) \longrightarrow N(0, 10^6) \longrightarrow \pi(\theta) \propto 1.$$

As the proper prior becomes more diffuse, its posterior approaches the flat-prior posterior. But finite diffuse and improper priors remain mathematically distinct objects.

9.4 Question 13.1 — Bernoulli Jeffreys Prior

Exact question from the assignment:

Suppose

$$Y \mid p \sim \text{Bernoulli}(p).$$

- (a) Write the log-likelihood.
- (b) Compute the first derivative with respect to p .
- (c) Compute the second derivative.
- (d) Derive the Fisher information

$$I(p) = -\mathbb{E} \left[\frac{\partial^2}{\partial p^2} \log f(Y \mid p) \right].$$

- (e) Derive Jeffreys prior

$$\pi_J(p) \propto \sqrt{I(p)}.$$

- (f) Identify the corresponding Beta distribution.
- (g) Compare this prior with the uniform prior

$$\text{Beta}(1, 1).$$

- (h) Explain why the two priors represent different notions of “noninformativeness.”

For a model with parameter θ , Fisher information can be written as

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X \mid \theta) \right].$$

Under regularity conditions it is also

$$I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X \mid \theta) \right)^2 \right].$$

Jeffreys’ prior is defined by

$$\pi_J(\theta) \propto \sqrt{I(\theta)}.$$

Its key motivation is invariance under smooth one-to-one reparameterization.

For a Bernoulli parameter p ,

$$Y \mid p \sim \text{Bernoulli}(p),$$

the resulting Jeffreys prior is

$$\pi_J(p) \propto p^{-1/2}(1-p)^{-1/2},$$

which is the

$$\text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

distribution.

Step 1: Log-likelihood.

For $Y = y \in \{0, 1\}$,

$$f(y | p) = p^y(1-p)^{1-y}.$$

Taking logarithms,

$$\ell(p) = y \log p + (1-y) \log(1-p).$$

Step 2: First derivative.

Differentiate:

$$\frac{\partial \ell}{\partial p} = \frac{y}{p} - \frac{1-y}{1-p}.$$

Thus

$$\ell'(p) = \frac{y}{p} - \frac{1-y}{1-p}.$$

Step 3: Second derivative.

Differentiate once more:

$$\frac{\partial^2 \ell}{\partial p^2} = -\frac{y}{p^2} - \frac{1-y}{(1-p)^2}.$$

Hence

$$\ell''(p) = -\frac{y}{p^2} - \frac{1-y}{(1-p)^2}.$$

Step 4: Fisher information.

By definition,

$$I(p) = -\mathbb{E}_p[\ell''(p)].$$

Since

$$\mathbb{E}_p[Y] = p, \quad \mathbb{E}_p[1-Y] = 1-p,$$

we get

$$I(p) = -\left[-\frac{p}{p^2} - \frac{1-p}{(1-p)^2}\right].$$

Therefore

$$I(p) = \frac{1}{p} + \frac{1}{1-p}.$$

Combine the fractions:

$$\frac{1}{p} + \frac{1}{1-p} = \frac{1-p+p}{p(1-p)} = \frac{1}{p(1-p)}.$$

Thus

$$I(p) = \frac{1}{p(1-p)}.$$

Step 5: Jeffreys prior.

Use

$$\pi_J(p) \propto \sqrt{I(p)}.$$

Therefore

$$\pi_J(p) \propto \sqrt{\frac{1}{p(1-p)}} = p^{-1/2}(1-p)^{-1/2}.$$

Hence

$$\pi_J(p) \propto p^{-1/2}(1-p)^{-1/2}.$$

Step 6: Identify the Beta distribution.

A Beta(a, b) density has kernel

$$p^{a-1}(1-p)^{b-1}.$$

We need

$$a-1 = -\frac{1}{2}, \quad b-1 = -\frac{1}{2}.$$

Thus

$$a = b = \frac{1}{2}.$$

Therefore

$$p \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right).$$

Step 7: Compare with the uniform prior.

The uniform prior is

$$\text{Beta}(1, 1),$$

whose density is constant.

Jeffreys' prior is

$$\text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right),$$

whose density is

$$\propto \frac{1}{\sqrt{p(1-p)}}.$$

This density places relatively more mass near $p = 0$ and $p = 1$ than the uniform prior.

Step 8: Different meanings of “noninformative”.

The uniform prior says

$$\pi(p) = \text{constant},$$

which is simple in the p -coordinate.

Jeffreys’ prior instead uses the geometry of the statistical model:

$$\pi_J(p) \propto \sqrt{I(p)}.$$

Thus it is constructed to be invariant under smooth reparameterization.

So the two priors encode different principles:

uniform in the parameter $\neq$ invariant objective prior.
--

9.5 Question 13.2 — Flat Priors Are Not Invariant

Exact question from the assignment:

Suppose

$$p \in (0, 1)$$

and define the log-odds parameter

$$\phi = \log \frac{p}{1-p}.$$

(a) Derive

$$p = \frac{e^\phi}{1 + e^\phi}.$$

(b) Compute

$$\frac{dp}{d\phi}.$$

(c) Suppose

$$\pi(p) \propto 1.$$

Derive the induced prior density for ϕ .

(d) Suppose instead

$$\pi(\phi) \propto 1.$$

Derive the induced prior density for p .

(e) Show explicitly that flatness is not preserved under transformation.

(f) Explain how Jeffreys prior addresses this issue.

Suppose a parameter is transformed by a smooth one-to-one map

$$\phi = h(p).$$

Densities transform by the Jacobian:

$$\pi_\phi(\phi) = \pi_p(p) \left| \frac{dp}{d\phi} \right|.$$

Therefore a density that is constant in p need not remain constant in ϕ .

For the Bernoulli log-odds parameter

$$\phi = \log \frac{p}{1-p},$$

the inverse transformation is

$$p = \frac{e^\phi}{1 + e^\phi},$$

and

$$\frac{dp}{d\phi} = p(1 - p) = \frac{e^\phi}{(1 + e^\phi)^2}.$$

Jeffreys' prior avoids the basic inconsistency because the Fisher information transforms with the square of the Jacobian, so

$$\sqrt{I(\phi)} d\phi = \sqrt{I(p)} dp.$$

Step 1: Find p as a function of ϕ .

Starting from

$$\phi = \log \frac{p}{1 - p},$$

exponentiate:

$$e^\phi = \frac{p}{1 - p}.$$

Multiply by $1 - p$:

$$e^\phi(1 - p) = p.$$

Expand:

$$e^\phi - e^\phi p = p.$$

Move the p -terms together:

$$e^\phi = p(1 + e^\phi).$$

Therefore

$$\boxed{p = \frac{e^\phi}{1 + e^\phi}}.$$

Step 2: Compute $dp/d\phi$.

Differentiate:

$$p(\phi) = \frac{e^\phi}{1 + e^\phi}.$$

Using the quotient rule,

$$\frac{dp}{d\phi} = \frac{e^\phi(1 + e^\phi) - e^\phi e^\phi}{(1 + e^\phi)^2}.$$

The numerator simplifies:

$$e^\phi + e^{2\phi} - e^{2\phi} = e^\phi.$$

Hence

$$\boxed{\frac{dp}{d\phi} = \frac{e^\phi}{(1 + e^\phi)^2}}.$$

Because

$$p = \frac{e^\phi}{1 + e^\phi}, \quad 1 - p = \frac{1}{1 + e^\phi},$$

this is also

$$\boxed{\frac{dp}{d\phi} = p(1 - p)}.$$

Step 3: Start with a flat prior in p .

Suppose

$$\pi(p) \propto 1.$$

The induced density for ϕ is

$$\pi_\phi(\phi) = \pi(p(\phi)) \left| \frac{dp}{d\phi} \right|.$$

Since $\pi(p) \propto 1$,

$$\pi_\phi(\phi) \propto \left| \frac{dp}{d\phi} \right|.$$

Thus

$$\boxed{\pi_\phi(\phi) \propto \frac{e^\phi}{(1 + e^\phi)^2}}.$$

This is not constant.

Step 4: Start with a flat prior in ϕ .

Suppose instead

$$\pi(\phi) \propto 1.$$

We want the induced density for p . Since

$$\phi = \log \frac{p}{1 - p},$$

differentiate:

$$\frac{d\phi}{dp} = \frac{1}{p} + \frac{1}{1 - p} = \frac{1}{p(1 - p)}.$$

Therefore

$$\pi_p(p) = \pi_\phi(\phi(p)) \left| \frac{d\phi}{dp} \right| \propto \frac{1}{p(1 - p)}.$$

Hence

$$\boxed{\pi_p(p) \propto \frac{1}{p(1 - p)}}.$$

Step 5: Show flatness is not preserved.

A flat density in p gives

$$\pi_\phi(\phi) \propto p(1 - p),$$

which is not flat in ϕ .

A flat density in ϕ gives

$$\pi_p(p) \propto \frac{1}{p(1-p)},$$

which is not flat in p .

Thus

“flat” depends on the chosen parameterization.

Step 6: How does Jeffreys’ prior solve this?

For one Bernoulli observation,

$$I(p) = \frac{1}{p(1-p)}.$$

Hence

$$\pi_J(p) \propto \sqrt{I(p)} = [p(1-p)]^{-1/2}.$$

The Jeffreys prior in the transformed coordinate is obtained from the transformed Fisher information. Because

$$I(\phi) = I(p) \left(\frac{dp}{d\phi} \right)^2,$$

we have

$$\sqrt{I(\phi)} = \sqrt{I(p)} \left| \frac{dp}{d\phi} \right|.$$

Therefore

$$\pi_J(\phi) d\phi \propto \sqrt{I(\phi)} d\phi = \sqrt{I(p)} dp \propto \pi_J(p) dp.$$

So the same prior measure is obtained in either coordinate.

Jeffreys’ prior is invariant under reparameterization.

9.6 Question 13.3 — Jeffreys Prior for a Normal Mean

Exact question from the assignment:

Suppose

$$Y \mid \mu \sim N(\mu, \sigma^2),$$

where σ^2 is known.

- (a) Derive the Fisher information for μ .
- (b) Obtain Jeffreys prior.
- (c) Determine whether the prior is proper.
- (d) Compare it with the commonly used flat prior

$$\pi(\mu) \propto 1.$$

- (e) Explain why they coincide in this location model.
- (f) Now consider a transformed parameter

$$\phi = a\mu + b, \quad a \neq 0.$$

Show how Jeffreys prior transforms.

- (g) Discuss why location models behave differently from bounded-parameter models such as the Bernoulli model.

For

$$Y \mid \mu \sim N(\mu, \sigma^2),$$

with σ^2 known, the Jeffreys prior is

$$\pi_J(\mu) \propto \sqrt{I(\mu)},$$

where

$$I(\mu) = -\mathbb{E}_\mu \left[\frac{\partial^2}{\partial \mu^2} \log f(Y \mid \mu) \right].$$

For a Normal location model,

$$I(\mu) = \frac{1}{\sigma^2},$$

which is constant in μ . Hence

$$\pi_J(\mu) \propto 1.$$

This is the usual flat prior on the real line, which is improper.

Under an affine transformation

$$\phi = a\mu + b, \quad a \neq 0,$$

the Jacobian is constant:

$$\left| \frac{d\mu}{d\phi} \right| = \frac{1}{|a|}.$$

Thus a flat prior remains flat up to a multiplicative constant.

Step 1: Fisher information.

The Normal density is

$$f(y | \mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}.$$

Therefore

$$\ell(\mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y - \mu)^2}{2\sigma^2}.$$

Differentiate:

$$\frac{\partial \ell}{\partial \mu} = \frac{y - \mu}{\sigma^2}.$$

Differentiate again:

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{1}{\sigma^2}.$$

Hence

$$I(\mu) = -\mathbb{E}_\mu \left[-\frac{1}{\sigma^2} \right] = \boxed{\frac{1}{\sigma^2}}.$$

Step 2: Jeffreys prior.

By definition,

$$\pi_J(\mu) \propto \sqrt{I(\mu)} = \sqrt{\frac{1}{\sigma^2}}.$$

Since σ^2 is known, this is a constant with respect to μ . Therefore

$$\boxed{\pi_J(\mu) \propto 1}.$$

Step 3: Is the prior proper?

The parameter space is all of $\mathbb{R}$. Thus

$$\int_{-\infty}^{\infty} 1 \, d\mu = \infty.$$

Hence

$$\boxed{\pi_J(\mu) \text{ is improper.}}$$

Step 4: Compare with the flat prior.

The commonly used flat prior is

$$\pi(\mu) \propto 1.$$

We have just derived

$$\pi_J(\mu) \propto 1.$$

Therefore

$$\pi_J(\mu) = \text{flat prior, up to proportionality.}$$

Step 5: Why do they coincide?

The Fisher information does not depend on μ :

$$I(\mu) = 1/\sigma^2.$$

Therefore its square root is constant, so Jeffreys' rule produces a flat density.

This is a hallmark of a location family: shifting μ does not change the local statistical difficulty of estimating it.

Step 6: Affine reparameterization.

Let

$$\phi = a\mu + b, \quad a \neq 0.$$

Then

$$\mu = \frac{\phi - b}{a},$$

so

$$\left| \frac{d\mu}{d\phi} \right| = \frac{1}{|a|}.$$

Starting from

$$\pi_J(\mu) \propto 1,$$

the transformed density is

$$\pi_J(\phi) \propto \pi_J(\mu(\phi)) \left| \frac{d\mu}{d\phi} \right| = 1 \cdot \frac{1}{|a|}.$$

Since $1/|a|$ is constant in ϕ ,

$$\pi_J(\phi) \propto 1.$$

Thus Jeffreys' prior remains flat up to its irrelevant multiplicative constant.

Step 7: Why is this different from the Bernoulli parameter?

The Normal mean takes values on the unbounded location space

$$\mathbb{R}.$$

Its information is constant, so Jeffreys' density is flat, and the integral diverges.

The Bernoulli parameter lies in the bounded interval

$$(0, 1),$$

and its information is

$$I(p) = \frac{1}{p(1-p)},$$

so Jeffreys' density becomes

$$p^{-1/2}(1-p)^{-1/2},$$

which is integrable on $(0, 1)$.

Thus the difference is not that Jeffreys' principle changes; the parameter space and information geometry are different.

9.7 Question 14.1 — KL Divergence for Bernoulli Models

Exact question from the assignment:

Let

$$P = \text{Bernoulli}(p), \quad Q = \text{Bernoulli}(q).$$

(a) Derive

$$D_{\text{KL}}(P\|Q).$$

(b) Show that

$$D_{\text{KL}}(P\|Q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

(c) Verify that

$$D_{\text{KL}}(P\|P) = 0.$$

(d) Explain why

$$D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)$$

in general.

(e) Let

$$q = p + \delta.$$

Use a Taylor expansion to obtain the second-order approximation of the KL divergence for small δ .

(f) Show how the Fisher information

$$I(p) = \frac{1}{p(1-p)}$$

appears in the approximation.

For two distributions P and Q with P absolutely continuous with respect to Q , the Kullback–Leibler divergence is

$$D_{\text{KL}}(P\|Q) = \mathbb{E}_P \left[\log \frac{p(X)}{q(X)} \right].$$

Important facts are:

$$D_{\text{KL}}(P\|Q) \geq 0,$$

with equality if $P = Q$ almost everywhere, and generally

$$D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P).$$

For a one-parameter regular model,

$$D_{\text{KL}}(f(\cdot | \theta) \| f(\cdot | \theta + \Delta)) = \frac{1}{2}I(\theta)\Delta^2 + o(\Delta^2),$$

so Fisher information is the local quadratic curvature of KL divergence.

Let

$$P = \text{Bernoulli}(p), \quad Q = \text{Bernoulli}(q).$$

Step 1: Derive the KL divergence.

The Bernoulli pmfs are

$$P(1) = p, \quad P(0) = 1 - p,$$

and

$$Q(1) = q, \quad Q(0) = 1 - q.$$

Therefore

$$D_{\text{KL}}(P \| Q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Hence

$$D_{\text{KL}}(P \| Q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Step 2: Verify the requested formula.

The expression already obtained is exactly

$$D_{\text{KL}}(P \| Q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

Step 3: Check $D_{\text{KL}}(P \| P) = 0$.

Set $q = p$:

$$D_{\text{KL}}(P \| P) = p \log \frac{p}{p} + (1 - p) \log \frac{1 - p}{1 - p}.$$

Since

$$\log 1 = 0,$$

we obtain

$$D_{\text{KL}}(P \| P) = 0.$$

Step 4: Why is KL generally asymmetric?

Interchanging p and q gives

$$D_{\text{KL}}(Q \| P) = q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p}.$$

There is no algebraic identity forcing this expression to equal

$$p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

Hence, in general,

$$\boxed{D_{\text{KL}}(P\|Q) \neq D_{\text{KL}}(Q\|P)}.$$

KL divergence is therefore not a distance in the ordinary symmetric metric sense.

Step 5: Set $q = p + \delta$.

Define

$$D(\delta) = p \log \frac{p}{p+\delta} + (1-p) \log \frac{1-p}{1-p-\delta}.$$

We now expand around $\delta = 0$.

Differentiate:

$$D'(\delta) = -\frac{p}{p+\delta} + \frac{1-p}{1-p-\delta}.$$

At $\delta = 0$,

$$D'(0) = -1 + 1 = 0.$$

Differentiate again:

$$D''(\delta) = \frac{p}{(p+\delta)^2} + \frac{1-p}{(1-p-\delta)^2}.$$

Therefore

$$D''(0) = \frac{1}{p} + \frac{1}{1-p}.$$

Combine:

$$\frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

Taylor's theorem gives

$$D(\delta) = D(0) + D'(0)\delta + \frac{1}{2}D''(0)\delta^2 + O(\delta^3).$$

Since

$$D(0) = 0, \quad D'(0) = 0,$$

we obtain

$$\boxed{D(\delta) = \frac{\delta^2}{2p(1-p)} + O(\delta^3)}.$$

Step 6: Identify Fisher information.

For one Bernoulli observation,

$$I(p) = \frac{1}{p(1-p)}.$$

Thus

$$D(\delta) = \frac{1}{2}I(p)\delta^2 + O(\delta^3).$$

Equivalently,

$$D_{\text{KL}}(\text{Bernoulli}(p) \parallel \text{Bernoulli}(p + \delta)) = \frac{1}{2}I(p)\delta^2 + o(\delta^2).$$

Interpretation

Fisher information measures how quickly the model distribution changes when p is perturbed. Large $I(p)$ means that even a small parameter change produces a larger KL divergence, so nearby parameter values are more statistically distinguishable.

9.8 Question 14.2 — Local Distinguishability

Exact question from the assignment:

Suppose

$$Y \mid \theta \sim f(y \mid \theta).$$

Define

$$D(\theta, \theta + d\theta) = D_{\text{KL}}(f(\cdot \mid \theta) \parallel f(\cdot \mid \theta + d\theta)).$$

(a) Write the log-density expansion

$$\log f(y \mid \theta + d\theta)$$

up to second order.

(b) Take expectation under

$$Y \sim f(\cdot \mid \theta).$$

(c) Show why the first-order term vanishes.

(d) Use the definition of Fisher information to obtain a quadratic approximation involving

$$I(\theta)(d\theta)^2.$$

(e) Explain why Fisher information can be interpreted as a local measure of distinguishability between nearby parameter values.

(f) Explain how this interpretation motivates

$$\pi_J(\theta) \propto \sqrt{I(\theta)}.$$

For a regular parametric model

$$Y \sim f(y \mid \theta),$$

define

$$D(\theta, \theta + d\theta) = D_{\text{KL}}(f(\cdot \mid \theta) \parallel f(\cdot \mid \theta + d\theta)).$$

If

$$\ell(\theta) = \log f(Y \mid \theta),$$

then

$$\ell(\theta + d\theta) = \ell(\theta) + \ell'(\theta)d\theta + \frac{1}{2}\ell''(\theta)(d\theta)^2 + o((d\theta)^2).$$

For a regular model,

$$\mathbb{E}_\theta[\ell'(\theta)] = 0$$

and

$$I(\theta) = -\mathbb{E}_\theta[\ell''(\theta)].$$

Therefore

$$D(\theta, \theta + d\theta) = \frac{1}{2}I(\theta)(d\theta)^2 + o((d\theta)^2).$$

This shows that Fisher information is the local curvature of KL divergence. Jeffreys' prior

$$\pi_J(\theta) \propto \sqrt{I(\theta)}$$

is therefore flat with respect to the Fisher–Rao distance

$$ds^2 = I(\theta)d\theta^2.$$

Step 1: Second-order expansion of the log-density.

Let

$$\ell(\theta) = \log f(y \mid \theta).$$

Taylor expansion around θ gives

$$\ell(\theta + d\theta) = \ell(\theta) + \ell'(\theta)d\theta + \frac{1}{2}\ell''(\theta)(d\theta)^2 + o((d\theta)^2).$$

Equivalently,

$$\log f(y \mid \theta + d\theta) = \log f(y \mid \theta) + \frac{\partial \log f}{\partial \theta} d\theta + \frac{1}{2} \frac{\partial^2 \log f}{\partial \theta^2} (d\theta)^2 + o((d\theta)^2).$$

Step 2: Insert the expansion into KL divergence.

By definition,

$$D(\theta, \theta + d\theta) = \mathbb{E}_\theta [\ell(\theta) - \ell(\theta + d\theta)].$$

Substitute the Taylor expansion:

$$D = -\mathbb{E}_\theta[\ell'(\theta)]d\theta - \frac{1}{2}\mathbb{E}_\theta[\ell''(\theta)](d\theta)^2 + o((d\theta)^2).$$

Step 3: Why does the first-order term vanish?

For a regular density,

$$\int f(y \mid \theta) dy = 1.$$

Differentiate with respect to θ :

$$\int \frac{\partial f(y | \theta)}{\partial \theta} dy = 0.$$

Since

$$\frac{\partial f}{\partial \theta} = f \frac{\partial \log f}{\partial \theta},$$

we obtain

$$\mathbb{E}_\theta \left[\frac{\partial \log f(Y | \theta)}{\partial \theta} \right] = 0.$$

Thus

$$\boxed{\mathbb{E}_\theta[\ell'(\theta)] = 0}.$$

Therefore the first-order contribution to KL divergence disappears.

Step 4: Use Fisher information.

By definition,

$$I(\theta) = -\mathbb{E}_\theta[\ell''(\theta)].$$

Hence

$$-\frac{1}{2} \mathbb{E}_\theta[\ell''(\theta)](d\theta)^2 = \frac{1}{2} I(\theta)(d\theta)^2.$$

Therefore

$$\boxed{D(\theta, \theta + d\theta) = \frac{1}{2} I(\theta)(d\theta)^2 + o((d\theta)^2)}.$$

Step 5: Interpretation as local distinguishability.

Suppose $I(\theta)$ is large. Then, even for a very small change $d\theta$,

$$D(\theta, \theta + d\theta) \approx \frac{1}{2} I(\theta)(d\theta)^2$$

is relatively large.

That means the distributions

$$f(\cdot | \theta) \quad \text{and} \quad f(\cdot | \theta + d\theta)$$

are easier to distinguish statistically.

Conversely, if $I(\theta)$ is small, nearby parameter values generate distributions that are locally harder to tell apart.

Thus

$$\boxed{I(\theta) \text{ measures local statistical distinguishability.}}$$

Step 6: Why does this motivate Jeffreys' prior?

The local squared distance is

$$ds^2 = I(\theta)d\theta^2.$$

Hence the corresponding infinitesimal distance is

$$ds = \sqrt{I(\theta)} |d\theta|.$$

A prior that is uniform with respect to this intrinsic statistical distance should satisfy

$$\pi_J(\theta) d\theta \propto ds.$$

Therefore

$$\boxed{\pi_J(\theta) \propto \sqrt{I(\theta)}}.$$

This gives a geometric interpretation of Jeffreys' rule: it is flat in statistical distance, not necessarily flat in an arbitrary coordinate.

9.9 Question 14.3 — KL Divergence and Posterior Consistency

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} f(\cdot \mid \theta_0).$$

Consider two parameter values

$$\theta_0 \quad \text{and} \quad \theta_1.$$

Define the log-likelihood ratio

$$L_n = \log \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_0)}.$$

(a) Show that

$$L_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_0)}.$$

(b) Divide by n and apply the law of large numbers.

(c) Show that the limiting expectation can be written as

$$-D_{\text{KL}}(f(\cdot \mid \theta_0) \parallel f(\cdot \mid \theta_1)).$$

(d) Assuming the KL divergence is strictly positive whenever

$$\theta_1 \neq \theta_0,$$

show that

$$L_n \rightarrow -\infty$$

at an approximately linear rate.

(e) Deduce that the likelihood ratio in favor of θ_1 tends to zero.

(f) Explain how this argument provides intuition for posterior concentration around the true parameter.

(g) Discuss the assumptions required before claiming full Bayesian consistency.

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} f(\cdot \mid \theta_0),$$

so that θ_0 is the true parameter. Compare θ_0 with an alternative θ_1 .

The likelihood ratio is

$$R_n = \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_0)}.$$

Because of conditional independence,

$$\log R_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_0)}.$$

The strong law of large numbers gives

$$\frac{1}{n} \log R_n \rightarrow \mathbb{E}_{\theta_0} \left[\log \frac{f(Y \mid \theta_1)}{f(Y \mid \theta_0)} \right].$$

But

$$D_{\text{KL}}(f(\cdot \mid \theta_0) \parallel f(\cdot \mid \theta_1)) = \mathbb{E}_{\theta_0} \left[\log \frac{f(Y \mid \theta_0)}{f(Y \mid \theta_1)} \right].$$

Therefore

$$\frac{1}{n} \log R_n \rightarrow -D_{\text{KL}}(f(\cdot \mid \theta_0) \parallel f(\cdot \mid \theta_1)).$$

If the KL divergence is strictly positive for $\theta_1 \neq \theta_0$, the log-likelihood ratio decreases linearly and $R_n \rightarrow 0$. Posterior odds equal prior odds times this likelihood ratio, so the posterior odds of θ_1 also vanish.

Step 1: Expand the log-likelihood ratio.

By definition,

$$L_n = \log \frac{p(Y_1, \dots, Y_n \mid \theta_1)}{p(Y_1, \dots, Y_n \mid \theta_0)}.$$

Conditional independence gives

$$p(Y_1, \dots, Y_n \mid \theta) = \prod_{i=1}^n f(Y_i \mid \theta).$$

Hence

$$L_n = \log \frac{\prod_{i=1}^n f(Y_i \mid \theta_1)}{\prod_{i=1}^n f(Y_i \mid \theta_0)}.$$

Use

$$\log \prod_i a_i = \sum_i \log a_i :$$

$$L_n = \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_0)}.$$

Step 2: Divide by n and apply the law of large numbers.

We have

$$\frac{L_n}{n} = \frac{1}{n} \sum_{i=1}^n \log \frac{f(Y_i \mid \theta_1)}{f(Y_i \mid \theta_0)}.$$

Under the true model θ_0 , the summands are i.i.d. Therefore the law of large numbers gives

$$\frac{L_n}{n} \longrightarrow \mathbb{E}_{\theta_0} \left[\log \frac{f(Y | \theta_1)}{f(Y | \theta_0)} \right].$$

Step 3: Express the limit through KL divergence.

By definition,

$$D_{\text{KL}}(f_{\theta_0} \| f_{\theta_1}) = \mathbb{E}_{\theta_0} \left[\log \frac{f(Y | \theta_0)}{f(Y | \theta_1)} \right].$$

Multiply inside the logarithm by reversing the ratio:

$$\log \frac{f(Y | \theta_1)}{f(Y | \theta_0)} = -\log \frac{f(Y | \theta_0)}{f(Y | \theta_1)}.$$

Hence

$$\frac{L_n}{n} \rightarrow -D_{\text{KL}}(f_{\theta_0} \| f_{\theta_1}).$$

Step 4: Show the approximately linear decay.

Assume

$$\theta_1 \neq \theta_0$$

and identifiability implies

$$D_{\text{KL}}(f_{\theta_0} \| f_{\theta_1}) > 0.$$

Let

$$D = D_{\text{KL}}(f_{\theta_0} \| f_{\theta_1}) > 0.$$

Then

$$\frac{L_n}{n} \rightarrow -D.$$

Thus, for large n ,

$$L_n \approx -nD.$$

Hence

$$L_n \rightarrow -\infty \quad \text{at an approximately linear rate.}$$

Step 5: Deduce the likelihood-ratio behavior.

Since

$$R_n = e^{L_n},$$

and

$$L_n \rightarrow -\infty,$$

we obtain

$$R_n \rightarrow 0.$$

More precisely,

$$R_n \approx e^{-nD},$$

so the rejection of the wrong parameter value occurs exponentially fast.

Step 6: Connect this to posterior concentration.

Posterior odds satisfy

$$\frac{P(\theta_1 | Y_{1:n})}{P(\theta_0 | Y_{1:n})} = \frac{P(\theta_1)}{P(\theta_0)} R_n.$$

The prior-odds factor is fixed, whereas

$$R_n \rightarrow 0.$$

Therefore

$$\boxed{\frac{P(\theta_1 | Y_{1:n})}{P(\theta_0 | Y_{1:n})} \rightarrow 0}.$$

So, asymptotically, the posterior favors the true parameter over the incorrect alternative.

The deeper intuition is that each observation contributes a small expected amount of evidence against the wrong model, and those contributions add. KL divergence determines the average rate at which the accumulated evidence grows.

Step 7: What is needed for full Bayesian consistency?

The two-point argument is powerful, but it is not by itself a complete proof of consistency in every continuous or multi-parameter model. One needs appropriate assumptions, including:

the statistical model is correctly specified,
the parameter is identifiable,
the relevant expectations exist,
the prior assigns positive mass near the true parameter,
and the alternative parameter space is controlled sufficiently for
posterior mass outside neighborhoods of the truth to vanish.

Thus the KL argument gives the core mechanism:

wrong model $\xrightarrow{\text{KL evidence}}$ exponentially worse likelihood $\xrightarrow{\text{Bayes}}$ posterior rejection.

Chapter 10

Integrated Challenge Problems

10.1 Integrated Challenge 1 — A Complete Beta–Bernoulli Analysis

Exact question from the assignment:

A new online platform believes that the probability θ that a user clicks an advertisement is unknown.

Historical data suggest

$$\theta \sim \text{Beta}(4, 16).$$

A new experiment observes

$$n = 50$$

users and

$$z = 15$$

clicks.

- (a) Write the full probability model.
- (b) Derive the posterior distribution.
- (c) Compute the posterior mean and posterior mode.
- (d) Derive the posterior variance.
- (e) Compare the posterior mean with the MLE.
- (f) Derive the posterior predictive probability of a click from the next user.
- (g) Derive the predictive distribution for the number of clicks among the next 10 users.
- (h) Explain how you would perform a posterior predictive check.
- (i) Compare the result with an analysis using the uniform prior.
- (j) Discuss whether the prior information remains important after observing 50 users.

For

$$Y_1, \dots, Y_n \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta), \quad \theta \sim \text{Beta}(\alpha, \beta),$$

the likelihood is

$$L(\theta) \propto \theta^z (1 - \theta)^{n-z}, \quad z = \sum_{i=1}^n Y_i.$$

Multiplying by the Beta prior kernel,

$$\pi(\theta \mid y) \propto \theta^{\alpha+z-1} (1 - \theta)^{\beta+n-z-1},$$

so

$$\boxed{\theta \mid y \sim \text{Beta}(\alpha + z, \beta + n - z)}.$$

For $X \sim \text{Beta}(a, b)$,

$$E(X) = \frac{a}{a+b},$$

$$\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)},$$

and, when $a, b > 1$,

$$\text{Mode}(X) = \frac{a-1}{a+b-2}.$$

For the next Bernoulli trial,

$$P(Y_{n+1} = 1 \mid y) = E[\theta \mid y].$$

For m future observations, if

$$K = \sum_{j=1}^m Y_{n+j},$$

then integrating over the Beta posterior gives the Beta–Binomial distribution:

$$P(K = k \mid y) = \binom{m}{k} \frac{B(a+k, b+m-k)}{B(a, b)}.$$

Here

$$\alpha = 4, \quad \beta = 16, \quad n = 50, \quad z = 15.$$

Step 1: Write the full probability model.

The hierarchical model is

$$\theta \sim \text{Beta}(4, 16), \quad Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

The joint distribution is

$$p(\theta, y_{1:50}) = p(\theta) \prod_{i=1}^{50} p(y_i \mid \theta).$$

Step 2: Derive the posterior.

The observed number of successes is $z = 15$, so the number of failures is

$$50 - 15 = 35.$$

The prior has kernel

$$\pi(\theta) \propto \theta^{4-1} (1-\theta)^{16-1}.$$

The likelihood has kernel

$$L(\theta) \propto \theta^{15} (1-\theta)^{35}.$$

Multiply:

$$\pi(\theta | y) \propto \theta^{3+15}(1 - \theta)^{15+35}.$$

Thus

$$\pi(\theta | y) \propto \theta^{18}(1 - \theta)^{50}.$$

Match this with

$$\theta^{a-1}(1 - \theta)^{b-1}.$$

Therefore

$$a = 19, \quad b = 51.$$

Hence

$$\boxed{\theta | y \sim \text{Beta}(19, 51)}.$$

Step 3: Posterior mean and mode.

For a $\text{Beta}(19, 51)$,

$$E[\theta | y] = \frac{19}{19 + 51} = \frac{19}{70}.$$

Thus

$$\boxed{E[\theta | y] = \frac{19}{70} \approx 0.27143}.$$

Because $19, 51 > 1$, the posterior mode is

$$\text{Mode}(\theta | y) = \frac{19 - 1}{19 + 51 - 2} = \frac{18}{68} = \frac{9}{34}.$$

Therefore

$$\boxed{\text{Mode}(\theta | y) = \frac{9}{34} \approx 0.26471}.$$

Step 4: Posterior variance.

Use

$$\text{Var}(\theta | y) = \frac{ab}{(a + b)^2(a + b + 1)}.$$

With $a = 19, b = 51$,

$$\text{Var}(\theta | y) = \frac{19 \cdot 51}{70^2 \cdot 71}.$$

Therefore

$$\boxed{\text{Var}(\theta | y) = \frac{969}{348100} \approx 0.00278528}.$$

Step 5: Compare posterior mean with MLE.

The MLE is the observed success fraction:

$$\hat{\theta}_{\text{MLE}} = \frac{z}{n} = \frac{15}{50} = 0.3.$$

The posterior mean is

$$0.27143.$$

Thus

$$\boxed{\text{posterior mean} < \text{MLE}}$$

because the prior mean is

$$\frac{4}{4 + 16} = 0.2,$$

which pulls the posterior downward from 0.3.

Indeed,

$$\frac{19}{70} = \frac{20}{70}(0.2) + \frac{50}{70}(0.3)$$

after writing the prior effective weight as $\alpha + \beta = 20$. Hence the posterior is a weighted compromise between prior information and data information.

Step 6: Posterior predictive probability for one new user.

For a future Bernoulli observation,

$$P(Y_{51} = 1 \mid y) = E[\theta \mid y].$$

Therefore

$$\boxed{P(Y_{51} = 1 \mid y) = \frac{19}{70} \approx 0.27143}.$$

Step 7: Predictive distribution for the next 10 users.

Let

$$K = \sum_{j=1}^{10} Y_{50+j}.$$

Conditional on θ ,

$$K \mid \theta \sim \text{Binomial}(10, \theta).$$

The posterior for θ is Beta(19, 51). Integrating θ out,

$$K \mid y \sim \text{Beta-Binomial}(10, 19, 51).$$

Hence

$$\boxed{P(K = k \mid y) = \binom{10}{k} \frac{B(19 + k, 61 - k)}{B(19, 51)}, \quad k = 0, \dots, 10.}$$

The predictive mean number of future clicks is

$$E[K \mid y] = 10 E[\theta \mid y] = 10 \cdot \frac{19}{70} = \frac{19}{7} \approx 2.714.$$

Thus

$$\boxed{E[K \mid y] \approx 2.714}.$$

Step 8: Posterior predictive check.

A natural check is to generate replicated datasets from the posterior predictive distribution and compare them with the observed data.

For example:

$$\theta^{(b)} \sim \text{Beta}(19, 51),$$

then

$$Z^{\text{rep},(b)} \sim \text{Binomial}(50, \theta^{(b)}).$$

Compare the observed

$$z_{\text{obs}} = 15$$

with the simulated distribution of Z^{rep} .

One may examine the proportion of replicated datasets with a discrepancy statistic at least as extreme as the observed one. If the observed data look very unusual relative to the posterior predictive distribution, that suggests the Bernoulli model may be inadequate.

Step 9: Compare with the uniform prior.

Under

$$\theta \sim \text{Beta}(1, 1),$$

the posterior would be

$$\theta \mid y \sim \text{Beta}(16, 36).$$

Its posterior mean is

$$\frac{16}{52} = \frac{4}{13} \approx 0.30769.$$

Thus the uniform prior gives a posterior mean much closer to the MLE 0.3, whereas the historical prior $\text{Beta}(4, 16)$ pulls strongly toward 0.2.

The corresponding posterior variance is

$$\frac{16 \cdot 36}{52^2 \cdot 53} \approx 0.004019.$$

This is larger than the variance 0.002785 under the historical prior, so the historical prior is both informative about the center and stronger in concentration.

Step 10: Does the prior still matter after 50 observations?

Yes, but less than it would with a small sample.

The prior mean is 0.2, while the sample proportion is 0.3. The posterior mean 0.2714 remains visibly below 0.3, so the prior has not disappeared.

However, as n grows, the data weight grows while the fixed prior parameters remain

constant. Therefore the prior influence eventually vanishes:

$$\hat{\theta}_{\text{Bayes}} \rightarrow \frac{z}{n}.$$

10.2 Integrated Challenge 2 — Competing Normal Models

Exact question from the assignment:

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} N(\theta, 1).$$

You are comparing

$$H_0 : \theta = 0$$

against

$$H_1 : \theta = 1.$$

The prior probabilities are

$$P(H_0) = 0.9, \quad P(H_1) = 0.1.$$

Suppose

$$n = 20, \quad \bar{Y} = 0.7.$$

(a) Derive the likelihood ratio

$$\frac{p(Y | H_1)}{p(Y | H_0)}.$$

(b) Compute the posterior odds.

(c) Compute the posterior probabilities.

(d) Determine whether the data overcome the strong prior preference for H_0 .

(e) Derive the expression for the log-likelihood ratio for general n and $\bar{Y}$.

(f) Assume the true parameter is $\theta = 1$. Use the law of large numbers to explain what happens to the posterior odds as $n \rightarrow \infty$.

(g) Interpret the limiting behavior using KL divergence.

Suppose

$$Y_1, \dots, Y_n \stackrel{\text{ind}}{\sim} N(\theta, 1)$$

and compare two point hypotheses

$$H_0 : \theta = 0, \quad H_1 : \theta = 1.$$

Posterior odds satisfy

$$\frac{P(H_1 | Y)}{P(H_0 | Y)} = \frac{P(H_1)}{P(H_0)} \frac{p(Y | H_1)}{p(Y | H_0)}.$$

For equal-variance Normal models,

$$\frac{p(Y | H_1)}{p(Y | H_0)} = \exp \left\{ n\bar{Y} - \frac{n}{2} \right\} = \exp \left\{ n \left(\bar{Y} - \frac{1}{2} \right) \right\}.$$

Thus the log-likelihood ratio is

$$\log BF_{10} = n \left(\bar{Y} - \frac{1}{2} \right).$$

Under the true model H_1 , $\bar{Y} \rightarrow 1$, so

$$\frac{1}{n} \log BF_{10} \rightarrow \frac{1}{2}.$$

This is also

$$D_{\text{KL}}(N(1, 1) \| N(0, 1)) = \frac{1}{2}.$$

We are given

$$\begin{aligned} P(H_0) &= 0.9, & P(H_1) &= 0.1, \\ n &= 20, & \bar{Y} &= 0.7. \end{aligned}$$

Step 1: Derive the likelihood ratio.

Under H_0 ,

$$f(y_i | H_0) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{y_i^2}{2} \right\}.$$

Under H_1 ,

$$f(y_i | H_1) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{(y_i - 1)^2}{2} \right\}.$$

Therefore

$$\frac{p(Y | H_1)}{p(Y | H_0)} = \prod_{i=1}^n \exp \left\{ -\frac{(Y_i - 1)^2 - Y_i^2}{2} \right\}.$$

Expand:

$$(Y_i - 1)^2 - Y_i^2 = Y_i^2 - 2Y_i + 1 - Y_i^2 = -2Y_i + 1.$$

Hence

$$-\frac{(Y_i - 1)^2 - Y_i^2}{2} = Y_i - \frac{1}{2}.$$

Therefore

$$BF_{10} = \exp \left\{ \sum_{i=1}^n \left(Y_i - \frac{1}{2} \right) \right\}.$$

Since

$$\sum_{i=1}^n Y_i = n\bar{Y},$$

we get

$$BF_{10} = \exp \left\{ n \left(\bar{Y} - \frac{1}{2} \right) \right\}.$$

Step 2: Numerical likelihood ratio.

Here

$$n = 20, \quad \bar{Y} = 0.7.$$

Thus

$$n \left(\bar{Y} - \frac{1}{2} \right) = 20(0.2) = 4.$$

Therefore

$$BF_{10} = e^4 \approx 54.598.$$

Step 3: Posterior odds.

The prior odds are

$$\frac{P(H_1)}{P(H_0)} = \frac{0.1}{0.9} = \frac{1}{9}.$$

Hence

$$\frac{P(H_1 | Y)}{P(H_0 | Y)} = \frac{1}{9} e^4.$$

Numerically,

$$\text{posterior odds} \approx 6.06646.$$

Step 4: Posterior probabilities.

Let

$$O_{10} = 6.06646.$$

Then

$$P(H_1 | Y) = \frac{O_{10}}{1 + O_{10}}.$$

Therefore

$$P(H_1 | Y) \approx \frac{6.06646}{7.06646} \approx 0.85849.$$

Hence

$$P(H_1 | Y) \approx 0.8585$$

and

$$P(H_0 | Y) \approx 0.1415.$$

Step 5: Do the data overcome the prior preference for H_0 ?

Initially,

$$P(H_0) = 0.9$$

and

$$P(H_1) = 0.1.$$

Thus H_0 has a strong prior advantage.

To overcome the prior in the sense of obtaining posterior odds greater than 1, we need

$$\frac{1}{9}BF_{10} > 1,$$

so

$$BF_{10} > 9.$$

The observed Bayes factor is

$$e^4 \approx 54.6 > 9.$$

Therefore the data do overcome the initial prior preference and produce posterior probability greater than 1/2 for H_1 .

Yes, the data overcome the prior preference for H_0 .

Step 6: General log-likelihood ratio.

From the derivation,

$$BF_{10} = \exp \left\{ n \left(\bar{Y} - \frac{1}{2} \right) \right\}.$$

Taking logarithms,

$$\log BF_{10} = n \left(\bar{Y} - \frac{1}{2} \right).$$

Step 7: Assume the true parameter is $\theta = 1$.

Under H_1 ,

$$Y_i \stackrel{\text{ind}}{\sim} N(1, 1).$$

By the law of large numbers,

$$\bar{Y} \rightarrow 1$$

almost surely.

Therefore

$$\frac{1}{n} \log BF_{10} = \bar{Y} - \frac{1}{2} \rightarrow \frac{1}{2}.$$

Hence

$$\log BF_{10} \sim \frac{n}{2},$$

so

$$BF_{10} \rightarrow \infty$$

exponentially fast.

The posterior odds are

$$\frac{1}{9}BF_{10},$$

and the fixed factor $1/9$ becomes negligible relative to exponential growth. Thus

$$\frac{P(H_1 | Y)}{P(H_0 | Y)} \rightarrow \infty$$

and consequently

$$P(H_1 | Y) \rightarrow 1.$$

Step 8: Interpret the limit using KL divergence.

For equal-variance Normals,

$$D_{\text{KL}}(N(\mu_1, 1) \| N(\mu_0, 1)) = \frac{(\mu_1 - \mu_0)^2}{2}.$$

Here

$$\mu_1 = 1, \quad \mu_0 = 0,$$

so

$$D_{\text{KL}}(N(1, 1) \| N(0, 1)) = \frac{1}{2}.$$

This is exactly the limiting per-observation log Bayes factor:

$$\frac{1}{n} \log BF_{10} \rightarrow \frac{1}{2}.$$

Thus the KL divergence gives the asymptotic rate at which evidence accumulates in favor of the true model:

$$\boxed{\text{KL divergence} = \text{asymptotic evidence rate.}}$$

10.3 Integrated Challenge 3 — From Exchangeability to Prediction

Exact question from the assignment:

Suppose

$$Y_1, Y_2, \dots$$

is assumed to be an infinitely exchangeable binary sequence.

By de Finetti's theorem, assume that there exists a latent random variable θ such that

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

Further suppose

$$\theta \sim \text{Beta}(\alpha, \beta).$$

After observing

$$n$$

observations with

$$z$$

successes:

(a) Derive the joint marginal distribution of the observed sequence.

(b) Show that the sequence is exchangeable.

(c) Derive the posterior distribution of θ .

(d) Derive

$$P(Y_{n+1} = 1 \mid Y_1, \dots, Y_n).$$

(e) Derive

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid Y_1, \dots, Y_n).$$

(f) Show that the future observations remain dependent after integrating out θ .

(g) Compute the posterior predictive covariance between

$$Y_{n+1}$$

and

$$Y_{n+2}.$$

(h) Explain how this single model simultaneously illustrates:

- exchangeability,

- conditional independence,
- marginal dependence,
- Bayesian updating,
- posterior prediction,
- conjugacy.

For an infinitely exchangeable binary sequence, de Finetti's theorem gives a latent parameter θ such that

$$Y_i \mid \theta \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta).$$

If

$$\theta \sim \text{Beta}(\alpha, \beta),$$

then, for $z = \sum_{i=1}^n y_i$,

$$p(y_{1:n} \mid \theta) = \theta^z (1 - \theta)^{n-z},$$

and integrating out θ ,

$$p(y_{1:n}) = \frac{B(\alpha + z, \beta + n - z)}{B(\alpha, \beta)}.$$

Because this depends on $y_{1:n}$ only through the number z of successes, it is invariant under permutation of the observations, hence exchangeable.

Beta–Bernoulli conjugacy gives

$$\theta \mid y_{1:n} \sim \text{Beta}(\alpha + z, \beta + n - z).$$

For the future observations,

$$P(Y_{n+1} = 1 \mid y) = E[\theta \mid y],$$

and

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid y) = E[\theta^2 \mid y].$$

For a $\text{Beta}(a, b)$ random variable,

$$E[\theta] = \frac{a}{a + b},$$

$$E[\theta^2] = \frac{a(a + 1)}{(a + b)(a + b + 1)},$$

and

$$\text{Var}(\theta) = \frac{ab}{(a + b)^2(a + b + 1)}.$$

Let

$$a = \alpha + z, \quad b = \beta + n - z.$$

Then a and b are the posterior Beta parameters.

Step 1: Joint marginal distribution of the observed sequence.

Conditional on θ ,

$$p(y_{1:n} \mid \theta) = \theta^z (1 - \theta)^{n-z}.$$

The Beta prior has density

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

Therefore

$$p(y_{1:n}) = \int_0^1 p(y_{1:n} \mid \theta) p(\theta) d\theta.$$

Substitute:

$$p(y_{1:n}) = \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha+z-1} (1 - \theta)^{\beta+n-z-1} d\theta.$$

The integral is the Beta function:

$$B(\alpha + z, \beta + n - z).$$

Hence

$$p(y_{1:n}) = \frac{B(\alpha + z, \beta + n - z)}{B(\alpha, \beta)}.$$

Step 2: Show exchangeability.

The expression above depends on the data only through

$$z = \sum_{i=1}^n y_i.$$

If the observations are permuted, the sum z does not change. Therefore

$$p(y_1, \dots, y_n) = p(y_{\pi(1)}, \dots, y_{\pi(n)})$$

for every permutation π .

Thus

$$\text{the sequence is exchangeable.}$$

Step 3: Derive the posterior distribution.

By Bayes' rule,

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta).$$

Hence

$$p(\theta \mid y) \propto \theta^z (1 - \theta)^{n-z} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

Collect exponents:

$$p(\theta \mid y) \propto \theta^{\alpha+z-1} (1 - \theta)^{\beta+n-z-1}.$$

Therefore

$$\theta \mid y \sim \text{Beta}(a, b) = \text{Beta}(\alpha + z, \beta + n - z).$$

Step 4: Predict the next observation.

By conditional independence,

$$P(Y_{n+1} = 1 \mid \theta, y) = \theta.$$

Average over the posterior:

$$P(Y_{n+1} = 1 \mid y) = \int_0^1 \theta p(\theta \mid y) d\theta = E[\theta \mid y].$$

For the $\text{Beta}(a, b)$ posterior,

$$E[\theta \mid y] = \frac{a}{a + b}.$$

Since

$$a + b = \alpha + \beta + n,$$

we obtain

$$P(Y_{n+1} = 1 \mid y) = \frac{\alpha + z}{\alpha + \beta + n}.$$

Step 5: Probability that both Y_{n+1} and Y_{n+2} equal 1.

Given θ ,

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid \theta, y) = \theta^2,$$

because the future observations are conditionally independent.

Therefore

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid y) = E[\theta^2 \mid y].$$

For $\text{Beta}(a, b)$,

$$E[\theta^2] = \frac{a(a+1)}{(a+b)(a+b+1)}.$$

Thus

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid y) = \frac{(\alpha + z)(\alpha + z + 1)}{(\alpha + \beta + n)(\alpha + \beta + n + 1)}.$$

Step 6: Show the future observations remain dependent.

If they were independent marginally given y , we would have

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid y) = P(Y_{n+1} = 1 \mid y)P(Y_{n+2} = 1 \mid y).$$

But

$$P(Y_{n+1} = 1, Y_{n+2} = 1 \mid y) = E[\theta^2 \mid y],$$

whereas

$$P(Y_{n+1} = 1 \mid y)P(Y_{n+2} = 1 \mid y) = \{E[\theta \mid y]\}^2.$$

Their difference is

$$E[\theta^2 \mid y] - \{E[\theta \mid y]\}^2 = \text{Var}(\theta \mid y).$$

For a non-degenerate Beta posterior,

$$\text{Var}(\theta \mid y) > 0.$$

Therefore

$$Y_{n+1} \not\perp Y_{n+2} \mid y.$$

Step 7: Compute the posterior predictive covariance.

Because Y is Bernoulli conditional on θ ,

$$E[Y_{n+1} \mid \theta, y] = \theta.$$

For $i \neq j$, conditional covariance is zero:

$$\text{Cov}(Y_{n+1}, Y_{n+2} \mid \theta, y) = 0.$$

The total covariance identity gives

$$\text{Cov}(Y_{n+1}, Y_{n+2} \mid y) = \text{Cov}(E[Y_{n+1} \mid \theta, y], E[Y_{n+2} \mid \theta, y] \mid y).$$

Hence

$$\text{Cov}(Y_{n+1}, Y_{n+2} \mid y) = \text{Var}(\theta \mid y).$$

For Beta(a, b),

$$\text{Cov}(Y_{n+1}, Y_{n+2} \mid y) = \frac{ab}{(a+b)^2(a+b+1)}$$

where

$$a = \alpha + z, \quad b = \beta + n - z.$$

Equivalently,

$$\text{Cov}(Y_{n+1}, Y_{n+2} \mid y) = \frac{(\alpha + z)(\beta + n - z)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}.$$

Step 8: Explain the six concepts illustrated simultaneously.

The same hierarchical model gives:

Exchangeability:	$p(y_{1:n})$ depends only on z .
Conditional independence:	$Y_i \stackrel{\text{ind}}{\sim} Y_j \mid \theta$.
Marginal dependence:	$\text{Cov}(Y_i, Y_j) = \text{Var}(\theta)$ in the relevant posterior/prior law.
Bayesian updating:	$p(\theta \mid y) \propto p(\theta)p(y \mid \theta)$.
Posterior prediction:	$P(Y_{n+1} = 1 \mid y) = \text{E}[\theta \mid y]$.
Conjugacy:	Beta $\rightarrow$ Beta under Bernoulli data.

The conceptual chain is therefore

latent $\theta \longrightarrow$ conditional iid $\longrightarrow$ exchangeability + marginal dependence $\longrightarrow$ posterior updating $\longrightarrow$ p

Master Formula Sheet

Bayes' rule

$$\pi(\theta | x) \propto \pi(\theta)f(x | \theta).$$

Beta–Bernoulli

$$\theta \sim \text{Beta}(\alpha, \beta), \quad Z = \sum_{i=1}^n Y_i, \quad Y_i | \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

implies

$$\boxed{\theta | y \sim \text{Beta}(\alpha + Z, \beta + n - Z)}.$$

Moreover,

$$\begin{aligned} \mathbb{E}[\theta | y] &= \frac{\alpha + Z}{\alpha + \beta + n}, \\ \text{Var}(\theta | y) &= \frac{(\alpha + Z)(\beta + n - Z)}{(\alpha + \beta + n)^2(\alpha + \beta + n + 1)}. \end{aligned}$$

Normal–Normal

$$\theta \sim N(\mu_0, \tau_0^2), \quad Y_i | \theta \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$$

implies

$$\boxed{\theta | y \sim N(\mu_n, \tau_n^2)}$$

with

$$\begin{aligned} \frac{1}{\tau_n^2} &= \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \\ \mu_n &= \frac{\mu_0/\tau_0^2 + (n/\sigma^2)\bar{y}}{1/\tau_0^2 + n/\sigma^2}. \end{aligned}$$

Posterior prediction

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta)p(\theta | y) d\theta.$$

For the Normal predictive model,

$$\tilde{Y} | y \sim N(\mu_n, \tau_n^2 + \sigma^2).$$

Jeffreys prior

$$\pi_J(\theta) \propto \sqrt{I(\theta)}$$

with

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X | \theta) \right].$$

For Bernoulli p ,

$$I(p) = \frac{1}{p(1-p)}, \quad \pi_J(p) \propto p^{-1/2}(1-p)^{-1/2}.$$

KL/Fisher local expansion

$$D_{\text{KL}}(f(\cdot | \theta) \| f(\cdot | \theta + d\theta)) = \frac{1}{2} I(\theta) (d\theta)^2 + o((d\theta)^2).$$

Posterior odds

$$\frac{P(H_1 | y)}{P(H_0 | y)} = \frac{P(H_1) p(y | H_1)}{P(H_0) p(y | H_0)}.$$